When LLM Judges Fail: Robustness Enhancement in Multimodal and Large-Scale Citation Scenarios

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Large language models are increasingly deployed as automatic judges of multimodal and retrieval-augmented outputs, yet the field still lacks a coherent account of when and why these judges can be trusted. This review synthesizes the evidence on LLM-as-judge reliability and shows that trustworthiness is governed less by aggregate agreement with humans than by whether the judge has access to independently verifiable evidence. The problem is difficult because judges are evaluated through single consistency numbers that conceal structured failures: visual cues are processed far more weakly than text, citation accuracy degrades with reasoning depth rather than document volume, and adversarial prompts such as refusal suppression and translation attacks reliably shift verdicts. We organize the literature around four questions—modality gaps, robustness to noise and citation demands, adversarial vulnerability, and the validity of prevailing metrics—and read each through the same evidence-channel lens. Across benchmarks spanning stereotype-sensitive occupations and gender, multi-hop question answering, long-context retrieval, and jailbreak suites, judges reach 86–96% agreement on single-hop tasks but fall to 72–75% on truth-seeking ones, while attack success rates vary from 0.05 to 0.34 depending on technique. Accuracy, Pearson correlation, and agreement rate emerge as necessary but insufficient lower bounds. The strongest conclusion is that reliable judging requires evidence channels independent of the judged output, and that evaluation practice should report structured, calibration-aware diagnostics rather than a single consensus score.

Additional Key Words and Phrases: multi-agent collaboration, large language models, agent orchestration, evaluation benchmarks, agent security

1 INTRODUCTION

Large language models (LLMs) are increasingly deployed as automated evaluators—so-called LLM-as-a-Judge systems—across natural language generation, safety screening, scientific question answering, and agentic tool use [1, 2]. As these judges extend from text-only settings into multimodal pipelines that score images, audio, video, and interleaved documents, a growing body of evidence documents systematic failures: judges misrank, over- or under-score, and collapse under distribution shift. The central tension motivating this review is that judge reliability is typically reported through single aggregate agreement statistics, while the actual failure surface is multi-dimensional—spanning modality, evidence provenance, adversarial pressure, and metric design. A judge that appears reliable under one protocol can be brittle under another.

This tension is sharpened in multimodal scenarios. Vision-language judges exhibit prototypicality and social-cue biases that text-only judges do not [3], and cross-modal coupling can induce preference collapse in self-evolving evaluators [4]. At the same time, hallucinations in multimodal models—whether fabricated or distorted [5]—propagate into judge decisions, while hallucination benchmarks and detectors mature largely independently of judge evaluation. Existing surveys and benchmarks tend to treat judge bias, adversarial robustness, and hallucination as separate literatures; few connect them into a unified account of where multimodal judges fail and why.

We argue that the field's blind spots are structural rather than incidental. First, judges are often evaluated without the evidence they would need to verify a claim, so apparent robustness reflects protocol artifacts rather than genuine verification capacity [6]. Second, adversarial and persuasion attacks exploit judge priors in ways that aggregate accuracy hides. Third, the dominant metrics—accuracy, Pearson correlation, agreement rate—are necessary but not sufficient, because they conflate ranking ability with calibrated scoring and mask directional error structure. These three observations organize the four questions this review addresses: modality-dependent performance gaps, robustness to retrieval noise and evidence provenance, vulnerability under adversarial attack, and the adequacy of evaluation metrics.

Our contribution differs from prior surveys in three respects. We synthesize structured evidence across

these four questions rather than cataloguing benchmarks, and we derive each claim from reported quantitative findings. We further connect mechanistic accounts of judge bias [7, 8] with mitigation strategies to explain why some interventions transfer and others do not. Finally, we frame metric inadequacy as a first-class failure mode rather than a measurement detail.

The remainder of this review is organized around four contributions. (i) We characterize modality-dependent blind spots, showing that visual processing lags textual processing and that social-cue and prototypicality biases persist even when faithfulness improves [3, 9]. (ii) We show that robustness to retrieval noise is governed less by noise volume than by whether the judge holds independently verifiable evidence, with multi-hop consistency systematically below single-hop. (iii) We document adversarial fragility under prompt injection, jailbreak, and persuasion, and show that no single defense jointly optimizes robustness, benign utility, and cost. (iv) We analyze why accuracy, correlation, and agreement are necessary-but-insufficient, and identify calibration-aware and bias-bounded alternatives. Throughout, we draw on hallucination-detection and mitigation work to explain how judge failures compound with generative failures, and we highlight open problems in cross-lingual and culturally grounded judge reliability.

2 BACKGROUND AND PRELIMINARIES

The rapid adoption of large language models (LLMs) as automated evaluators—commonly termed LLM-as-a-Judge—has transformed how the research community assesses model outputs at scale. In this paradigm, a capable LLM is prompted to score, rank, or compare candidate responses, replacing or augmenting costly human annotation. Recent work has formalized this practice and begun to catalog its reliability boundaries: judges exhibit systematic biases, miscalibration, and sensitivity to prompt design. As evaluation pipelines extend from text to images, video, agent trajectories, and retrieval-grounded question answering, the failure surface of automated judging widens considerably. This section defines the core concepts that recur throughout the review and clarifies how they relate, providing the vocabulary needed for the thematic chapters that follow.

Having revealed how judging failures compound with generation failures and having identified open problems in the reliability of cross-lingual and culturally grounded judging, it is necessary to turn to the emergence and systematization of the LLM-as-a-Judge paradigm itself, to examine how it reshapes large-scale model evaluation, and to organize the research threads on reliability issues.

2.1 LLM-as-a-Judge and Multimodal Judging

We define an LLM judge as a language model (or vision-language model) that receives a task specification, a candidate output, and optionally a reference, and produces a judgment in the form of a score, preference, or label. When the judged content includes non-textual modalities, we refer to the evaluator as a multimodal judge. The concept has evolved from simple pairwise preference prompting to structured, rubric-driven evaluation protocols [2]. Multimodal judging introduces distinct challenges: judges must ground judgments in visual evidence, and their scoring behavior diverges from text-only settings. For instance, vision-language judges can reliably rank candidates yet fail to produce calibrated absolute scores, a task-dependent uncertainty that complicates metric aggregation [9]. Prompt optimization has been proposed to partially close this gap [10], while dedicated frameworks such as ScratchEval and FORGE illustrate domain-specific instantiations of multimodal evaluation [11]. We treat the judge as an evaluator whose reliability is itself an empirical property to be measured, not assumed.

2.2 Hallucination and Cross-Modal Failure

Hallucination denotes generated content that is unfaithful to the input or to verifiable knowledge. In multimodal settings, hallucinations manifest as fabricated objects, distorted spatial relations, or unsupported temporal claims in video [5]. Hallucination is not merely a generator defect; it propagates into evaluation, because a judge that misperceives visual evidence will issue judgments grounded in fabricated content. Architectural and training factors modulate hallucination robustness in large vision-language models [12], and mitigation strategies range from phase-wise self-rewarding to sharpness-aware erasure [13] and hallucination-aware in-context learning. Benchmarks such as KGHaluBench provide structured probes of knowledge breadth and depth [14]. We define cross-modal failure as the case where the judge's textual reasoning and visual evidence decouple, producing confident but unsupported verdicts; this coupling is itself a measurable phenomenon in self-evolving evaluators [4].

2.3 Judging Bias, Anchoring, and Context Dependence

Judging bias refers to systematic deviations from a normative standard that arise from factors irrelevant to output quality, such as answer position, verbosity, self-similarity, or stylistic polish. Anchoring describes the judge's over-reliance on an initial reference or retrieved context, such that judgments shift with the anchor rather than with the candidate's intrinsic merit; this is especially acute in retrieval-augmented generation (RAG) pipelines, where long contexts induce information forgetting and context dependence. Related biases include self- and other-label effects [15] and rationalization bias, wherein judges construct post-hoc justifications for predetermined verdicts. Bias-bounded evaluation offers a formal route to provable fairness constraints [16], and active preference learning provides a statistical treatment of biased judges.

2.4 Adversarial Vulnerability and Robustness

Adversarial vulnerability of a judge is its susceptibility to inputs crafted to manipulate the verdict, including prompt injection, adversarial images, and post-decision interaction attacks. Evidence suggests that judges are unreliable even in measuring safety and jailbreak robustness, behaving closer to a coin flip than a stable oracle. Defenses include lightweight adversarial frameworks such as StableJudge, bias-mitigating judge architectures [17], and extraction-resistant protocols. We distinguish robustness (stability under perturbation) from manipulability (directional susceptibility to adversarial steering), a distinction formalized in recent stress tests.

2.5 Benchmarks, Metrics, and Calibration

A benchmark is a standardized dataset and protocol for measuring evaluator behavior; examples span agentic tool-calling [18], multi-turn synthetic judging, multilingual and cross-trajectory reliability [19], culturally grounded multimodal evaluation, medical multimodal assessment [20], and safety-oriented QA. Metrics such as accuracy, agreement rate, and Pearson correlation quantify judge quality, but their validity depends on calibration—the alignment between judge scores and human perception or ground truth. Deterministic alternatives to LLM judging, such as GroundEval, illustrate a complementary design philosophy for stateful agent evaluation [21], while reference-based lightweight judges such as BERT-as-a-Judge offer efficient baselines [22]. We define calibration standardization as the process of validating and normalizing judge outputs against curated reference distributions so that scores are comparable across tasks, models, and modalities.

2.6 Relationships Among Concepts

These concepts form a causal chain rather than a list. Hallucination and perceptual error corrupt the evidence a judge reasons over; bias and anchoring distort how that evidence is weighted; adversarial manipulation targets both stages; and benchmarks with calibrated metrics are the instruments by which the resulting failures are detected and quantified. The blind spots emphasized in this review arise precisely at the intersections: a multimodal judge may perceive correctly yet anchor on a retrieved reference, or rank robustly yet score without calibration. The following chapters examine each link in this chain, beginning with the empirical anatomy of what a judge must be shown and how much [6].

3 RELATED WORK

The evaluation of large language models has shifted from reference-overlap metrics toward model-based judgment, and a growing body of surveys and empirical studies now maps this transition. Comprehensive overviews frame LLM-as-a-Judge as a general evaluation paradigm, cataloguing its application scenarios, prompting strategies, and known limitations. Alongside such broad syntheses, targeted surveys examine the reliability of judges in alignment tasks and document systematic weaknesses across preference-comparison settings [23]. A parallel strand of work interrogates robustness directly, arguing that judges cannot yet be trusted under distribution shift and adversarial pressure [24]. These surveys converge on a shared conclusion: judge quality is contingent, task-dependent, and poorly characterized outside text-only, general-purpose dialogue. The present review differs by organizing this literature around failure mechanisms rather than around application domains, and by extending the analysis to multimodal and retrieval-grounded settings that existing surveys treat only marginally.

After clarifying the empirical structure of the inputs required by evaluators, the focus of evaluation practice naturally shifts to the paradigm of modeled judgment itself: the transition from reference-overlap

metrics to LLM-as-a-Judge is not merely a replacement of measurement tools, but further implies that the judging subject, prompting strategies, and known limitations are re-systematically orchestrated.

The foundational methodology literature establishes both the promise and the protocol of model-based evaluation. Early work demonstrates that strong LLMs can approximate human preferences at scale, validating judge-based comparison on multi-turn benchmarks and crowd-sourced arenas [25]. Subsequent studies refine the reference condition under which judgments are rendered, showing that reference-guided verdicts improve the evaluation of free-form question answering [26]. A related line decomposes judgments into atomic and holistic components for reference-grounded support labeling, finding that prompt-controlled decomposition changes the reliability profile of the judge [27]. Work on multi-turn dialogue evaluation aggregates signals from multiple judges to stabilize conversational assessment [28]. More recent designs move beyond single-model prompting toward agentic and personalized judges, in which an agent plans, retrieves, and revises its own evaluation procedure, and toward automatic judge design tailored to specific generation applications. This review builds on these protocol-level insights but treats them as a baseline against which robustness under adversarial and multimodal conditions must be measured.

A second methodological stream treats the judge as an estimator whose bias must be calibrated or corrected rather than merely prompted. CalibraEval calibrates the prediction distribution of judges to mitigate selection bias, reframing judgment as a distributional estimation problem [29]. Contrastive decoding is repurposed to suppress score-range bias, showing that decoding-time interventions can shift judge outputs without retraining [30]. Assistant-guided mitigation addresses teacher preference bias, where a judge inherits the stylistic preferences of its own training signal. Bias-bounded evaluation pursues provably unbiased judges through formal constraints on the evaluation procedure. Complementary work evaluates and mitigates judge bias specifically in communication systems, where automated quality assessment directly affects downstream decisions. These approaches share a corrective orientation; the present review instead asks which biases survive correction once the input modality or the evidence base changes, and documents failures that calibration cannot absorb.

Bias in judging has also been studied comparatively against human annotators and across model families. A systematic comparison of human and LLM judges identifies overlapping but non-identical bias profiles, challenging the assumption that LLMs simply inherit human heuristics. Multi-agent judging is examined for whether additional perspectives amplify or resist prejudice, with results suggesting that aggregation does not monotonically reduce bias [31]. Shortcut bias is shown to operate silently, with judges relying on surface cues that they never acknowledge in their rationales [32]. In the code domain, judges are found to be sensitive to superficial features of programs rather than to functional correctness. Epistemic markers constitute a further confound: expressions of uncertainty in candidate answers systematically shift judge scores [33]. A causal framework for rationalization bias traces how judges fabricate post-hoc justifications for verdicts reached through spurious cues. This review draws on these findings to argue that bias taxonomies developed for text do not transfer cleanly to multimodal inputs, where visual salience introduces additional shortcut channels.

Robustness and adversarial evaluation form the most active empirical front. A comprehensive assessment of fifteen attacks and eight defenses reports that no single defense simultaneously optimizes robustness, benign utility, and cost, with re-tokenization achieving low residual attack success at severe utility loss [24]. The coin-flip finding extends this pessimism to safety evaluation, showing that judges fail to reliably measure adversarial robustness and that reported safety gains may be artifacts of judge noise. Adversarial persuasion demonstrates that judges can be talked into inflated scores through argumentative framing rather than factual content [34]. The Judge Reliability Harness provides reusable stress-testing infrastructure for constructing validation suites [35]. Meta-evaluation of safety judges shows that robustness conclusions depend heavily on the meta-metric chosen. Benchmarking adversarial robustness to bias elicitation couples scalable red-teaming with judge-based assessment, revealing that elicitation succeeds across model families. This review extends the threat model beyond text prompts to adversarial images and manipulated retrieval evidence, a combination that existing robustness studies do not jointly examine.

The limits of scalable evaluation have been formalized rather than merely observed. Theoretical analysis shows that LLM judges cannot beat twice the data at the frontier, bounding the accuracy gains available from model-based annotation. The no-free-labels argument demonstrates that judges without human grounding provide unreliable signals in high-stakes domains [36]. A methodological critique asks whether the community is assessing judges in the right way, noting that existing meta-evaluation protocols conflate agreement with validity [37]. Knowledge-driven failures in reference-conditioned QA judging show that judges misapply their own knowledge when a reference is present [38]. Scientific question answering provides a further testbed, where robust judging must contend with domain-specific correctness criteria. This review adopts the validity-versus-agreement distinction as a organizing principle and applies it to

settings where the reference itself is retrieved and therefore fallible.

Hallucination research supplies both a conceptual vocabulary and a set of detection techniques that inform judge reliability. Lifecycle-based surveys organize causes, detection, mitigation, and prevention of hallucination into a coherent pipeline [39]. Classification-oriented studies taxonomize hallucination types and pair each with mitigation strategies, while investigative work compares detection and mitigation methods empirically. A provocative position argues that hallucination is intrinsic to current training objectives and must be managed rather than eliminated [40]. Detection methods range from weakly supervised probing of internal activations [41] to fast Fourier analysis of hidden-layer temporal signals. Mitigation strategies include self-reflection [42], smoothed knowledge distillation [43], and adaptive retrieval augmentation [44]. This review connects this literature to judging by treating judge hallucination—fabricated rationales and unsupported verdicts—as a first-class failure mode rather than an auxiliary concern.

Retrieval-augmented generation introduces a distinct evidence regime in which the judge must assess answers against retrieved context that may itself be incomplete or noisy. Debate-augmented RAG shows that multi-agent debate substantially improves exact match on retrieval-insufficient subsets, raising scores on multi-hop benchmarks where the gold answer is absent from retrieved documents. Debate-driven experiments further examine how adversarial exchange affects both accuracy and hallucination rates. Multi-agent architectures are proposed as a general mitigation strategy for hallucination in deployed systems [45], and teaming arrangements among LLMs are shown to improve detection and mitigation [46]. Agentic settings raise the stakes: MIRAGE-Bench documents hallucinative actions in LLM agents and localizes where they originate [47]. This review argues that these retrieval and agentic scenarios expose judge failures that static, reference-based benchmarks cannot reveal, and that citation-level accuracy remains largely unmeasured.

Multimodal evaluation has developed its own benchmark tradition, much of it predating or running parallel to judge-based assessment. Crossmodal-3600 established massively multilingual image-captioning evaluation with human references [48], while fine-grained visual commonsense work probes whether models cross-reference image and text rather than relying on language priors [49]. Controlled diagnostic datasets such as ColorSwap isolate color and word-order competence [50]. MEGA-BENCH scales multimodal evaluation to over five hundred real-world tasks [51], and dynamic bootstrapping generates evaluation instances of flexible complexity. A systematic study clarifies the role that LLMs actually play inside multimodal benchmarks, distinguishing judging from scoring [52]. Specialized frameworks extend multimodal evaluation to podcast audio generation [53], block-based programming [54], and concept erasure in diffusion models [55]. This review differs from this benchmark-centric tradition by asking how judges behave when they are the evaluator of multimodal content, rather than how benchmarks measure multimodal models.

A smaller but growing literature places LLMs directly in the role of multimodal evaluators. FLEX-Judge shows that text-only reasoning can unlock zero-shot multimodal evaluation, decoupling judge capability from perceptual access [56]. Bi-level prompt optimization tunes multimodal judges jointly at the instruction and instance level. Vision-language model judges are found to rank candidates competently while failing to assign calibrated scores, with rank correlation on chart understanding reaching only moderate levels [9]. Tree-structured reasoning-process scoring evaluates faithfulness in multimodal settings by scoring intermediate reasoning steps. Prototypicality bias in automatic metrics reveals blind spots that persist when metrics are used as judge proxies [57]. Benchmarking vision-language models with social-cue news images pairs stimulus design with judge-based assessment [3]. This review synthesizes these findings into a claim that multimodal judges exhibit a rank-score dissociation, and that this dissociation is aggravated by adversarial visual input.

Safety-critical and domain-specific judging constitute a further application cluster with distinct validity requirements. Safety alignment evaluation in Chinese mental health dialogues illustrates how judges must operate where gold-standard answers are unavailable or ethically constrained [58]. Communication-system judging ties evaluation quality to operational consequences [59], while safety-judge meta-evaluation questions whether robustness claims survive changes in the evaluation metric. Scientific QA judging adds domain expertise as a confound [60], and materials-science verification demonstrates multi-stage checking of generated content. Agent evaluation motivates deterministic alternatives to judge-based scoring for stateful systems [1]. This review treats these domain studies as evidence that judge validity is task-relative, and uses them to motivate the cross-domain robustness comparison we conduct.

Hallucination in multimodal and embodied settings has received attention that is largely disconnected from the judging literature. Video-LLM work targets action-scene hallucination through disentangled spatial-temporal representations [61], and robust erasure methods address object hallucination in multimodal LLMs [13]. Agentic reasoning approaches such as Hydra improve adversarial robustness while reducing hallucination in vision-language models [62]. Visual-signal analysis detects hallucination in sign-

language translation, where fluent output may be unsupported by visual evidence [63]. A unified theoretical account links adversarial fragility and hallucination through an uncertainty principle [64]. Benchmarking work asks whether hallucination can be reframed as a measurable capability rather than a defect [65]. This review argues that these two literatures—multimodal hallucination and judge reliability—have developed in isolation, and that their intersection is precisely where judge failures in multimodal evaluation originate.

Benchmark and metric design for judging has matured into a distinct methodological concern. Hallucination benchmarks such as HalluLens provide standardized protocols for measuring fabrication [66], and knowledge-comparative benchmarks assess the breadth and depth of model knowledge [14]. Multitask, multilingual, multimodal evaluations of interactive models expose reasoning and hallucination behavior under controlled prompts. Consistency metrics such as agreement rate and Pearson correlation are widely reported, yet they establish only marginal proximity to a human reference and remain necessary but insufficient conditions for validity [9]. The distinction between binary proxy scores and calibrated judgments recurs across these efforts, and calibration-set availability emerges as a practical constraint on rigorous meta-evaluation [35]. This review adopts a multi-metric stance, reporting agreement alongside rank correlation and adversarial success rates, and argues that no single metric captures judge validity across modalities.

Adjacent evaluation traditions offer methodological lessons that the judge literature has not fully absorbed. Work on exposure bias and domain shift in neural machine translation shows how evaluation conditions interact with generation pathologies [67]. Multimodal evaluation has long confronted the problem of reference scarcity, from multilingual captioning [48] to no-reference image quality assessment via controllable list-wise ranking. Language-grounding resources such as Touchdown provide shareable evaluation environments for instruction following [68], and spatial-reasoning studies test whether LLMs can plan paths under prompt engineering. Multimodal style transfer via graph cuts and concept-bank automation illustrate how evaluation criteria are entangled with the generative objective. This review draws on these traditions to argue that judge design should inherit the reference-scarcity discipline of no-reference evaluation rather than assuming human references are available.

Finally, a set of application studies demonstrates the breadth of domains in which judge-based evaluation is now deployed, and the corresponding breadth of failure modes. Clinical multimodal evaluation of long-COVID patients and neurovascular function in Parkinson's disease shows how multimodal assessment protocols are validated in medical contexts. Assistive navigation systems for blind painters are validated through multimodal evaluation approaches, and ethnic tourism imaginaries are analyzed through multimodal evaluation of social media content [69]. Ideological drift in news-grounded question answering reveals directional hallucination that judges may fail to flag. These studies collectively show that evaluation validity is domain-contingent, and that judges trained or prompted on general dialogue may systematically misjudge specialized content. This review uses this evidence to justify a cross-domain robustness protocol rather than a single-domain benchmark.

Taken together, existing work covers four broad perspectives: general surveys of the judge paradigm [70, 23], bias and calibration methods [30, 16], adversarial and robustness assessments [24, 71], and multimodal benchmarks with emerging judge-based evaluators [51, 10]. Three blind spots persist. First, multimodal judging is studied through benchmarks but rarely through adversarial stress tests that manipulate visual evidence. Second, retrieval-grounded judging lacks metrics for citation accuracy and context robustness, despite evidence that debate and retrieval interventions change outcomes substantially [72]. Third, the hallucination and judge-reliability literatures remain disconnected, so fabricated rationales are documented but not systematically attributed. This review addresses these gaps by unifying adversarial, multimodal, and retrieval-grounded perspectives under a single validity framework, and by reporting rank-score dissociation and citation-level failures as primary evidence rather than as secondary observations.

- [70] — comprehensive survey framing the judge paradigm, its scenarios, and limitations.
- [24] — systematic assessment of fifteen attacks and eight defenses, showing no single defense dominates.
- [71] — demonstrates that judges fail to reliably measure adversarial robustness.
- [25] — foundational validation of judge-based evaluation against human preferences.
- [72] — debate-augmented RAG improving multi-hop QA under retrieval insufficiency.
- [9] — evidence that vision-language judges rank but do not calibrate scores.
- [56] — text-only reasoning enabling zero-shot multimodal evaluation.

- [73] — theoretical bound on judge-based scalable evaluation.
- [36] — argument that judges without human grounding yield unreliable labels.

4 SURVEY METHODOLOGY

Of the 116 screened publications, 81 are cited in the synthesis; the remaining 35 informed screening and background and are not individually cited.

Data Mapping. The corpus comprises 116 publications assembled through multi-source retrieval, with arXiv serving as the primary channel and open scholarly indexes used as fallback coverage; records were deduplicated by anchoring each entry to its DOI or equivalent persistent identifier. The review is organized around 4 research questions decomposed into 8 sub-questions, and the resulting evidence set was fixed prior to analysis so that no paper entered or left the corpus once coding began. This mapping defines both the provenance of the material and its boundaries: every claim in the synthesis traces back to one or more of the 116 anchored records, and the sub-question structure determines how those records are distributed across the analysis.

After pointing out that evaluation labels without a human grounding are unreliable and presenting the theoretical limits of scalable evaluation, the construction and organization of the corpus become the empirical precondition for testing these claims: the following explains the retrieval, deduplication, and coding process for the 116 references, as well as the review structure developed around 4 research questions and 8 sub-questions.

Data Refinement. Each of the 116 papers was coded into six evidence fields: problem addressed, method, datasets and benchmarks, metrics, limitations, and stated assumptions and constraints. Coding proceeded paper by paper, and cross-paper correspondences were recorded so that papers sharing a problem framing, a benchmark, or a methodological lineage were linked rather than treated as isolated entries. The refinement stage thus fixed the evidence in a comparable form before any interpretive work began: the six fields supply a uniform schema, and the cross-paper links supply the relational structure through which comparison, evolution, and trade-off analysis are later conducted.

Data Evaluation. Evaluation proceeds in three layers. First, an evidence matrix is constructed from the coded fields, aligning the 116 papers against the 8 sub-questions so that coverage and gaps are visible per sub-question. Second, each sub-question is analyzed in turn through comparison across papers, tracing of methodological evolution, and explicit treatment of trade-offs among competing approaches. Third, conclusions are decomposed into checkable claims, each annotated with the supporting papers, so that every assertion in the synthesis is traceable to specific anchored records rather than to the corpus as an undifferentiated whole. This three-layer protocol keeps the fixed evidence set and the derived claims separate, allowing the strength of support behind each claim to be inspected independently.

5 BLIND SPOTS IN MULTIMODAL EVALUATION AND THE SPECTRUM OF FAILURE MODES

After establishing a layered protocol in which each individual assertion can be traced back to an anchored record, the organization of evidence shifts from a static archive to a genealogical review aggregated by thesis: what follows unfolds the research landscape presented by the 116 literature items along thematic lines, so that the strength of support under each thesis can be compared horizontally within the same framework.

Having established the methodological apparatus for multimodal judgment, we now turn to what that apparatus systematically fails to see. The central tension of this chapter is that LLM judges exhibit a marked asymmetry between modalities: textual evidence is processed with comparative fluency, whereas visual cues—though capable of carrying social signals such as gender, race, and occupation—are frequently misread, and in some cases amplified rather than attenuated. We trace this asymmetry through the mechanisms by which language priors dominate decoding and suppress visual grounding, producing hallucinated descriptions that diverge from the image they purport to assess, and through the cross-modal interactions in which these failures compound. The chapter closes by following the consequences of such blind spots into the judging process itself, where anchoring on prior context and biased reference points begin to distort verdicts—the terrain the next chapter examines in RAG settings.

5.1 Judge Performance Gap under Visual Input and the Text-Dominated Evaluation Paradigm

Does an LLM judge that performs reliably on text preserve that reliability once images or video enter the input, and if not, where does the degradation originate? The evidence assembled for this chapter

indicates that the answer is negative in a specific and structured way: visual modality does not merely add noise to an otherwise stable textual evaluation capability, but introduces systematic biases and capability regressions that have no counterpart in text-only judging. At the same time, the dominant evaluation protocols remain text-dominated, so that the visual channel is often present in the input without being causally responsible for the verdict. The subsections below separate these two phenomena—genuine visual capability deficits and protocol-level neglect of the visual channel—and examine the quantitative evidence for each.

The most direct evidence of a visual-specific capability deficit comes from controlled colour–word composition tasks. On ColorSwap, mainstream vision-language models perform close to chance, with reported task accuracy of roughly 72% for GPT-4V and 87% for BLIP, figures that stand in sharp contrast to text-only evaluation accuracy of about 87% on the corresponding textual formulation. The gap is not a matter of general visual competence but of precise compositional binding: models that describe scenes fluently still fail to match colour terms to the correct objects under order-sensitive conditions, and fine-tuning yields only partial improvement. This pattern suggests that the deficit is architectural rather than attentional—an inability to maintain fine-grained cross-modal correspondence—and it implies that a judge relying on such representations cannot be trusted to adjudicate fine-grained visual attributes even when its textual judgements are sound.

A second and more consequential failure appears in the social domain, where visual cues are not merely misperceived but actively amplified. A systematic study of gender and occupation cues in news imagery finds that multimodal models frequently preserve faithfulness to the image while simultaneously strengthening stereotypes relative to the textual baseline. Occupation and gender are identified as the most sensitive cue categories, and the effect is directional rather than random: the visual channel pushes judgements toward the prototypical association instead of correcting it. This is qualitatively different from text-only bias, where stereotypes are largely carried by lexical choice and can be attenuated by prompt or calibration interventions. In the visual case the bias enters through the perceptual pathway, which existing text-side mitigation strategies do not reach.

Prototypicality bias provides a mechanistic account of how this amplification operates. On PROTOBIAS, vision-language judges are shown to prefer typical images over semantically correct but atypical ones, a preference that is nearly absent in text-only evaluation. The mechanism is plausibly one of representational priors: the judge's visual encoder assigns higher likelihood to canonical appearances, and this prior leaks into the verdict as an implicit quality signal. The consequence for evaluation practice is that a model can appear to reward correctness while in fact rewarding typicality, and the two criteria diverge precisely on the minority and edge cases that bias audits are designed to detect. This finding also helps explain why the stereotype amplification observed on news imagery is systematic rather than idiosyncratic.

The fragility of visual grounding is further exposed under adversarial latent conditions. On M-ErasureBench, concept removal rates exceed 70% under latent inversion, indicating that visually grounded judgements are unstable when the latent representation is perturbed [29]. The significance of this result lies in its asymmetry: text-driven erasure methods generalize poorly to the visual latent space, so a judge validated on textual concept removal offers little assurance about its behaviour on images. Read together with the ColorSwap and PROTOBIAS results, this points to a layered vulnerability—compositional binding, representational typicality, and latent instability—each of which operates through a different mechanism and none of which is captured by text-only validation.

Against this picture of visual-specific deficits stands a striking counter-result on the sufficiency of textual reasoning. FLEX-Judge, trained on only 1K textual reasoning examples, surpasses LLaVA-Critic on VL-RewardBench and MJ-Bench while showing no appreciable performance drop on TextVQA or OK-VQA. The comparison is instructive because the two systems differ in exactly the resource that a modality-centric view would consider essential: LLaVA-Critic depends on large-scale multimodal annotation, whereas FLEX-Judge relies on text-only reasoning data. That the latter generalizes across modalities suggests that much of what a multimodal judge needs is transferable reasoning skill rather than modality-specific perception, and it implies that scaling multimodal supervision is not the only—or necessarily the most efficient—route to cross-modal evaluation competence.

This counter-result, however, should not be read as evidence that visual input is being used well. The same body of work indicates that most multimodal evaluation protocols remain text-dominated, with judges relying primarily on textual representations and treating visual input as auxiliary rather than decision-relevant. The distinction matters: FLEX-Judge's success demonstrates that textual reasoning transfers, not that visual evidence is integrated. A judge can thus score well on multimodal benchmarks while effectively ignoring the image, which means benchmark performance and genuine visual grounding are partially decoupled. This decoupling is itself a failure mode, because it allows a system to pass multimodal

evaluation without possessing the capability the evaluation purports to measure.

Calibration evidence sharpens the contrast between the modalities. CalibraEval mitigates selection bias by adjusting the predicted distribution, reducing rank standard deviation on MT-Bench by roughly 10–15 points and improving accuracy by 2–5 points. These gains are achieved in the textual setting, where the judge's output distribution is directly observable and adjustable. The natural question is whether the same intervention transfers to visual judging, and the current evidence does not answer it: calibration has not been validated on visual modality at comparable scale, and the distributional assumptions underlying the method may not hold when the input includes perceptual uncertainty. The asymmetry in maturity between text-side and vision-side calibration is itself a gap in the literature.

The divergence in explanations for these blind spots deserves explicit treatment. One strand attributes the deficits to the visual modality itself, arguing that social cues and perceptual priors introduce biases absent from text [39]. Another locates the problem in evaluation protocol design, pointing to prototypicality preferences and text-dominated judging as artifacts of how benchmarks are constructed rather than of the modality per se. These accounts are not mutually exclusive—architectural constraints and protocol deviations plausibly interact—but the literature has not yet separated them empirically, and doing so would require benchmarks that hold protocol constant while varying modality, which current resources do not provide.

The boundaries of the available evidence reinforce this caution. The social-cue study rests on 1,343 image pairs drawn from static news imagery, leaving dynamic social context and interactive evaluation unexamined [39]. PROTOBIAS targets text-to-image evaluation and has not been extended to judge evaluation proper [42], while M-ErasureBench captures visual fragility only indirectly through concept removal rates [29]. Cross-lingual generalization is likewise unresolved: multilingual image-captioning benchmarks exist, but no study directly compares visual-modality judging against text-modality judging across languages, so whether the visual channel introduces language-specific bias remains an open question. These gaps are not incidental; they mark the boundary between what the current evidence establishes and what evaluation practice currently assumes.

Taken together, the evidence supports a two-part conclusion. First, visual modality introduces genuine and mechanistically distinct failure modes—compositional binding errors, stereotype amplification, prototypicality preference, and latent instability—that text-only validation cannot detect. Second, the dominant response to these failures has been to lean on textual reasoning, which transfers well across modalities but does so partly by sidelining the visual channel, producing judges that score well without grounding well. The blind spot is therefore double: a capability deficit in visual understanding and a protocol-level tendency to route around it. The next section turns from this diagnosis of modality-specific failure to the broader spectrum of failure modes that cut across modalities, examining how these deficits couple with errors in reasoning, calibration, and aggregation.

5.2 Language Prior Dominance and Cross-Modal Contamination in Judge Decisions

How do hallucination, visual misunderstanding, and cross-modal interaction failure actually manifest when an LLM Judge is asked to evaluate multimodal content? The evidence assembled here indicates that these are not three independent defects but a single coupled mechanism: linguistic priors dominate decoding, visual evidence is under-utilized, and bias originating in one modality propagates into judgments about another. This section traces that mechanism from its generative origin through its behavioural signature to its contamination effects, and closes by identifying where the evidence thins out.

The generative origin of judge-side hallucination lies in the conditional dominance of the language prior over visual evidence. When visual input is ambiguous or weakly grounded, the model completes the judgment from linguistic co-occurrence rather than from the image, producing outputs that read as plausible but are visually unfaithful. Work on tree-structured reasoning processes shows that faithfulness to the visual input degrades precisely at the steps where the reasoning chain is longest and the visual signal weakest. A stronger claim in the literature holds that hallucination is not a removable artifact but follows from basic limits of decidability in the underlying formal setting, implying that no amount of scale or instruction tuning will eliminate it entirely. For evaluation, this reframes hallucination from a defect to be filtered out into a baseline rate that any judge protocol must explicitly bound.

The behavioural signature of this prior dominance is a systematic dissociation between task-level accuracy and item-level consistency. Vision-language models can score highly on a composite reasoning task while giving severely inconsistent answers to the sub-questions that compose it, which indicates that the headline score is carried by shortcuts rather than by scene understanding. The ME benchmark operationalizes exactly this concern by decomposing evaluation into sub-questions probing visual scene, text,

and background knowledge separately, exposing inconsistencies that aggregate scores conceal. In contrast, benchmarks that report only end-task accuracy cannot distinguish a judge that reads the image from one that reads the prompt; the fine-grained decomposition is what makes the distinction observable at all.

A parallel line of evidence locates the problem in the evaluation protocol rather than in the model alone. PROTOCOLBENCH demonstrates that evaluation metrics can systematically reward representativeness over factual accuracy, so a judge optimized against such metrics is incentivized toward fluent, typical descriptions rather than correct ones. This is a protocol-level analogue of the language-prior effect: just as the model prefers likely text to verified perception, the metric prefers typical output to true output. The two biases compound, and a judge trained or prompted against a representativeness-rewarding metric will inherit the very shortcut the metric encodes.

Quantifying the resulting reliability gap makes the severity concrete. Across multimodal judging studies, exact-match agreement with human evaluation falls in the 32–34% range and Spearman correlation with human preference in the 0.30–0.46 range, far below what is routinely assumed when LLM judges are used as ground truth. Calibration analysis further shows that evaluation uncertainty varies sharply by task type, with interval coverage spanning roughly 40% to 70% depending on the task. The practical implication is that a judge's confidence is not transferable across task types: a protocol calibrated on one multimodal task may be badly miscalibrated on another, and the aggregate correlation figure hides this heterogeneity.

Cross-modal contamination is the third and least visible failure mode. Systematic study of evaluator preference coupling finds that a single strategy can absorb 48.4% of the effective weight while visually grounded strategies receive only 9.1%, meaning the judge's decision is dominated by a modality-agnostic policy rather than by visual evidence. This asymmetry explains why judges can appear stable and self-consistent while being blind: consistency is supplied by the dominant linguistic strategy, not by agreement with the image. The coupling is directional and asymmetric, and it is this asymmetry, rather than the mere presence of bias, that makes the failure hard to detect from output statistics alone.

Aggregation architecture modulates rather than removes these biases, and the direction of the modulation is counter-intuitive. Comparing debiasing strategies across position, verbosity, chain-of-thought, and conformity biases, multi-agent debate is found to amplify bias rather than attenuate it, whereas an LLM-as-Meta-Judge design resists it more effectively. The mechanism is plausibly conformist: debate gives each agent exposure to others' positions, and under a strong language prior the majority position converges on the fluent rather than the visually correct answer. Optimization-based recalibration of the predicted distribution offers a complementary route, reducing measured bias without altering the underlying architecture. The comparison matters because it shows that adding deliberation is not a neutral intervention; it can deepen the very coupling it is meant to break.

Benchmark coverage reveals where the field can and cannot see these failures. M-ErasureBench, the first comprehensive multimodal concept-erasure benchmark, spans textual prompts, learned embeddings, and inverse-inference latents, and reports that erasure fails in non-textual modalities with concept retention rates above 90%. This is a striking contrast with text-only settings, where erasure is comparatively effective: the modality that carries the most evaluation weight is the one where intervention is least reliable. Multilingual multimodal resources such as Crossmodal-3600, built with bilingual annotation to avoid translation artifacts, extend coverage along the language axis but not along the faithfulness axis [48]. Coverage is therefore uneven in a specific way: breadth across languages and tasks is growing, while depth of visual-fact verification remains thin.

Data contamination compounds the measurement problem and is unevenly distributed across modalities. Reported contamination rates reach 84.46% for image–text pairs and 33.13% for text alone in a widely used multilingual multimodal benchmark, a gap that indicates contamination enters preferentially through the visual channel. The consequence for judging is direct: a judge that has seen benchmark items during pretraining will score them by recall, and because the contamination is heavier on the visual side, the inflation is mistaken for visual competence. Contamination and prior dominance thus reinforce each other, and neither can be estimated reliably while the other is uncontrolled.

Domain-transfer evidence suggests the coupling is not confined to general-purpose benchmarks. Prospective multimodal evaluation in a clinical long-COVID setting shows that multimodal assessment pipelines can be constructed and audited end to end, yet the reliance on structured instruments leaves open how much of the judgment is driven by the image versus the accompanying text. Exposure-bias research in neural models supplies a complementary caution: models systematically over-weight features that were frequent during training, which in a multimodal judge translates into over-weighting the textual channel that dominates pretraining corpora. Together these results indicate that the language-prior effect observed on synthetic benchmarks has a plausible domain-general counterpart, though the direct evidence

in specialized domains remains indirect.

Two boundaries on this evidence deserve emphasis. First, several bias studies rest on small repeated samples, so the magnitude of cross-modal coupling is estimated with limited precision even where its direction is clear. Second, much of the reliability evidence derives from a single large judging benchmark, and generalization to real deployment conditions of greater complexity and diversity is not established. These are not reasons to discount the mechanism, which is supported by convergent evidence from generation, protocol, and aggregation studies, but they do limit how precisely its parameters can be stated.

Taken together, the evidence supports a single account: hallucination in multimodal judging originates in language-prior dominance, manifests as inconsistency between aggregate and item-level performance, and is amplified into cross-modal contamination by aggregation designs that reward conformity. The failure is therefore structural rather than incidental, and it is masked by benchmarks that measure task completion without verifying visual facts. This account sets up the broader question taken up next: how these modality-specific couplings interact with errors in reasoning, calibration, and aggregation to produce the full spectrum of judge failure.

6 JUDGING BIAS AND ANCHORING FAILURE IN RAG SCENARIOS

The modality coupling limitations exposed by visual fact-checking immediately give rise to a more general question: how do errors in the reasoning, calibration, and aggregation stages intertwine with them to jointly constitute the full spectrum of judgment failures?

Building on the preceding account of blind spots in multimodal evaluation, this chapter turns to how judging bias and anchoring failure arise when retrieval-augmented generation supplies the evidence on which a judge must rely. The central tension is that a judge's robustness depends less on the volume of retrieval noise than on whether it holds an evidence channel that can be independently verified rather than one derived from the very output under scrutiny. Accordingly, the chapter first examines citation accuracy and multi-hop noise to show how shared provenance between evidence and output produces anchoring, then traces how context-length limits, information forgetting, and dependence on retrieved context degrade judgment consistency. These threads converge on a picture of the judge as structurally vulnerable to manipulation of its evidence, which sets up the adversarial perspective developed in the next chapter.

6.1 Retrieval Noise and Citation Accuracy as Stress Tests for Judge Robustness

How robust is an LLM judge when the evidence it must evaluate is itself noisy, incomplete, or multi-hop in structure? This question sits at the intersection of retrieval-augmented generation and automatic evaluation, and the evidence assembled here suggests that judge reliability degrades not with the sheer volume of retrieved material but with the fidelity of the correspondence between evidence and claim. The findings below trace three connected stress dimensions: the irreparability of retrieval failure, the asymmetry between distracting and misleading evidence, and the systematic leniency that appears when citation anchors are absent.

A first and sobering result concerns the limits of mitigation under retrieval failure. On subsets where the gold answer is entirely absent from retrieved documents, debate-augmented RAG improves average EM from 3.60 to 5.04 on NQ, from 7.10 to 19.23 on 2WikiMultihopQA, and from 9.32 to 17.36 on HotpotQA, yet absolute EM remains below 20% in every case [72]. The mechanism is instructive: debate can redistribute probability mass toward correct candidates when partial cues survive in the context, but it cannot manufacture evidence that was never retrieved. Retrieval noise thus defines a failure region that deliberation narrows without closing, which implies that judge robustness must be assessed conditionally on retrieval recall rather than reported as a single aggregate.

The same study reveals that more deliberation is not monotonically better, a pattern that complicates any assumption that additional reasoning rounds strengthen judgment. Retrieval debate saturates early: 2WikiMultihopQA EM rises from 26.00 at zero retrieval iterations to 29.60 at two, with marginal gains thereafter, while response debate is non-monotonic, with TriviaQA EM falling from 62.40 at one response iteration to 60.20 at two [72]. In contrast to the retrieval-failure result, where added computation helped but could not suffice, here added computation actively harms once the useful signal is exhausted. The two findings together indicate that the marginal value of debate depends on whether unresolved uncertainty remains in the evidence, a condition that judges—and the systems that configure them—currently lack explicit machinery to detect.

A second stress dimension concerns what kind of noise actually threatens judgment. Counter to the

intuition that long contexts overwhelm evaluators, burying a genuine source document among ten irrelevant ones costs only 2–5 points of recall, and the most relevant 11% of source sentences retain 98% of full-source recall (0.908 versus 0.927) [6]. Plausible-but-wrong documents, by contrast, raise false positives from 12.4 to 20.9. Document count is therefore not the dominant constraint; provenance between evidence and output is. This distinction reframes the robustness problem: judges appear reasonably resistant to volume and markedly vulnerable to semantic near-misses, which is precisely the condition that multi-hop retrieval frequently produces.

The fragility induced by near-miss evidence extends to reference provision itself, where ostensibly helpful material can be worse than none. Supplying a correct reference summary alongside the article erodes 82% of the precision gain attributable to the article (balanced accuracy 0.769 to 0.618; false positives 31% to 70%), whereas a reference alone performs almost identically to blind judging (0.552 versus 0.565) [6]. A parallel pattern emerges in reference-guided verdict studies: high human agreement holds only when the judge can itself answer the question correctly and a correct reference is available, and a slightly erroneous reference is worse than an unrelated or absent one [36]. The convergence across independent designs is striking—partial or approximate grounding does not merely fail to help; it actively misleads.

If misleading evidence is the primary hazard, explicit citation requirements emerge as the most cost-effective countermeasure documented so far. Requiring judges to number source sentences and attach a citation to every identified defect reduces false positives from 3.36 to 2.88 per error-free answer at a cost of only 0.5 points of recall [6]. This is the lowest-cost precision gain reported in the relevant comparisons, and its logic aligns with the provenance finding above: forcing the judge to localize its claim in the evidence suppresses the plausibility-driven false positives that volume-based defenses fail to address. Citation accuracy, on this reading, is not merely an output-format preference but a structural constraint on judge reasoning.

Multi-hop structure compounds these difficulties, and the degradation is measurable in human agreement. Judge–human agreement falls systematically from single-hop to multi-hop settings: PA reaches 91%/92%/96% on HotpotQA and 86%/90%/90% on TriviaQA, but only 72%/75%/74% on TruthfulQA across Mistral-J, GPT-J, and Llama-J configurations [26]. The interpretation is that citation and reasoning-chain accuracy become harder to adjudicate as the number of inferential steps grows, since each hop introduces an opportunity for the judge to accept a locally plausible but globally unsupported link. This provides a mechanistic account of why the retrieval-failure results above are so severe on 2WikiMultihopQA and HotpotQA specifically: multi-hop tasks combine longer chains with more near-miss distractors, the two conditions under which judges are least reliable.

Absent citation anchors, judges drift toward leniency rather than skepticism, and the drift is large. Without a reference, judges over-credit correct-looking answers: accuracy is highest in the no-reference setting and declines once a reference is introduced, and the decision-flip rate from no-reference to reference-verified ranges from 0.09 to 0.85, far exceeding the 0.01–0.24 range for reference-verified to reference-cited transitions. The effect is not confined to English or to high-resource settings: on Telugu, Gemma3-27B accepts erroneous answers at a rate of 60% [74]. Combined with the earlier finding that a wrong reference is worse than none, this yields a non-monotonic reliability curve across grounding conditions—blind judges are generous, partially grounded judges are misled, and only correctly grounded judges approach human agreement.

These judge-side vulnerabilities do not stand in isolation; they interact with biases documented elsewhere in evaluation research. Selection bias in judge outputs can be calibrated at the prediction-distribution level [29], and multi-agent judging arrangements can amplify rather than attenuate prejudice depending on how perspectives are aggregated [31]. Adversarial probing further shows that judge verdicts are unreliable under distribution shift, with safety judgments behaving close to chance in some configurations [71], and that bias elicitation can be systematically benchmarked through judge-mediated assessment [75]. Read against the retrieval results, these findings suggest a common substrate: judges are sensitive to the surface form and framing of evidence rather than to its evidential status, whether the perturbation arrives as a near-miss document, a flawed reference, or an adversarially phrased prompt.

The lifecycle perspective on hallucination offers a unifying frame for these observations. If hallucination arises from causes spanning data, training, and inference, and if detection and mitigation must be matched to the originating stage [39], then a judge evaluating RAG output is effectively performing stage-attribution under partial observability. The empirical pattern—volume tolerated, near-misses harmful, citations helpful, multi-hop harder—is consistent with judges that can detect gross absence of support but struggle to verify fine-grained provenance. This account also explains why debate saturates: deliberation improves attribution only while distinguishable evidence remains, and collapses into noise once the evidence set is exhausted [72].

Taken together, the evidence supports a conditional conclusion. LLM judges are robust to retrieval volume but fragile to retrieval fidelity; explicit citation requirements and correct references improve reliability, while approximate references and near-miss distractors degrade it below the no-reference baseline; and multi-hop tasks concentrate both hazards. The practical implication is that robustness claims for judges should be reported jointly with retrieval recall, reference correctness, and hop count, since aggregate agreement figures conceal exactly the conditions under which judges fail. These anchoring effects—where the judge's verdict is pulled toward whatever evidence is most proximate rather than most probative—form the natural bridge to the next section, which examines how such anchoring manifests as a systematic bias in judge decisions rather than an incidental sensitivity to input quality.

6.2 Context-Window Saturation, Forgetting, and Reference Dependence as Coupled Consistency Failures

How do finite context windows, information forgetting, and reliance on retrieved context jointly undermine the consistency of LLM judges? The evidence assembled here indicates that these are not three independent nuisances but a single coupled mechanism: judges fail to exploit the guidance they are given, forget the early portions of long inputs, and, once reference material is removed or perturbed, fall back on their own priors rather than on the evidence supplied to them. This section traces that mechanism from the input-side constraint through the forgetting process to the reference-side dependence, and it closes by identifying where the coupling is strongest and where current mitigations only partially relieve it.

The first link in the chain is positional: a judge's use of its input is not uniform across the context window, so guidance placed early loses influence as the input grows. Evidence from the anatomy of judge blind spots shows that what a judge is shown, and how much of it, materially changes its verdict, implying that the effective context is considerably narrower than the nominal window [6]. The consequence for few-shot judging is direct: demonstrations supplied at the head of a long prompt are progressively discounted, so the same judge behaves differently across instances that differ only in prompt length rather than in task difficulty. This is a consistency failure in the strict sense, because the judge's decision rule drifts with an input property that is irrelevant to correctness. The mechanism is best understood as an attention-allocation effect rather than a capacity limit per se; the information remains present in the prompt but no longer carries proportionate weight in the verdict.

Forgetting compounds this positional effect, and the two are difficult to separate empirically. As inputs lengthen, judges lose reliable access to specific earlier content, which produces verdicts that waver between the retrieved context and the judge's own knowledge [39]. The lifecycle account of hallucination is instructive here because it frames forgetting and confabulation as two faces of the same retrieval failure: when the relevant span cannot be recovered, the model fills the gap with plausible content rather than signalling uncertainty [39]. Applied to judging, this means a long-context judge may not merely underweight early evidence but may substitute a reconstructed version of it, yielding confident but unfaithful verdicts. A natural question is whether truncation would help; the evidence suggests it trades one failure for another, since shortening the context restores attention to the remaining material at the cost of discarding evidence the judge was meant to weigh.

Reference dependence is the third and best-quantified link, and it is where the coupling with forgetting becomes visible. Without a reference answer, judges exhibit a pronounced over-crediting tendency, marking incorrect responses as correct because nothing in the input constrains them to do otherwise [74]. The same pattern appears in evaluations without human grounding, where judge scores diverge systematically from human labels once the anchoring provided by ground truth is absent [36]. Supplying a reference reverses the error direction rather than eliminating error: in reference-swap experiments, adding or altering reference information flips correct/incorrect decisions in a substantial fraction of cases, with reported accuracy losses in the range of thirty to forty percent. The interpretation is that the judge does not reliably condition on the reference at all; it conditions on a mixture of reference and prior, and the mixture weight shifts with context length and reference placement. Forgetting and reference dependence therefore reinforce each other, because a judge that has lost track of the reference span is functionally a judge without a reference.

Reference-guided designs partially relieve this dependence but do not remove it, and the comparison across reference regimes is instructive. Providing human references yields the highest agreement and the lowest bias, self-reference introduces self-enhancement favouring the judge's own outputs, and the no-reference regime is consistently the weakest [36, 74]. Reference-guided verdict frameworks attempt to make the reference do explicit work in the decision, improving faithfulness when the reference is present and well-formed [26]. However, the improvement is conditional: it depends on the reference being correctly aligned with the candidate response, which is precisely the condition that fails under swap or perturbation. Calibration-based approaches offer a complementary route by reshaping the judge's prediction distribution

rather than its input, mitigating selection bias without requiring additional reference material [29]. These two families are therefore addressing different links in the chain—input-side grounding versus output-side recalibration—and neither alone closes the loop.

Multi-agent judging, often proposed as a remedy for single-judge bias, interacts with all three links in a way that can amplify rather than attenuate inconsistency. When several judges deliberate, the information each agent contributes is itself subject to the same window and forgetting constraints, so early contributions lose weight as the debate transcript grows, and the aggregation can converge on a position that no individual agent would have reached from the full evidence [31]. Debate-augmented retrieval shows that structured disagreement can reduce hallucination when the debate is anchored to retrieved evidence [72], which suggests the amplification is conditional on whether the deliberation remains grounded. The contrast is therefore not between single and multi-agent judging as such, but between grounded and ungrounded aggregation: the former can correct individual forgetting, the latter compounds it.

The boundary conditions deserve emphasis, because the coupling is not uniform across tasks. Adversarial robustness evaluations show that judge reliability degrades sharply under distribution shift, with agreement approaching chance on some safety judgements [71], and bias-elicitation benchmarks demonstrate that judge-scored assessments are themselves sensitive to the elicitation procedure [75]. These findings indicate that the context and reference mechanisms described above are most damaging when the judged content is unusual relative to the judge's priors, since that is when the judge most needs the supplied evidence and is least able to use it. Conversely, on well-formed, in-distribution items with clean references, the same judges can appear highly consistent, which is why benchmark-level agreement figures can mask the underlying fragility.

Taken together, the evidence supports a single conclusion: context-window saturation, forgetting, and reference dependence form a coupled failure mode whose signature is verdict instability under input perturbations that should be irrelevant to correctness. The coupling explains why single-link mitigations—truncation, better references, calibration, or multi-agent aggregation—each produce partial gains without eliminating the underlying sensitivity. This raises the question of whether the residual instability is a property of how judges are prompted or of how they are evaluated, a distinction that the next section takes up by examining the metrics and protocols through which judge consistency is currently measured.

7 JUDGING VULNERABILITY AND DEFENSE UNDER ADVERSARIAL ATTACKS

Since it remains to be determined whether the residual instability originates from the prompting method or the evaluation method, the metrics and protocols for measuring judge consistency become the next object of investigation.

Building on the preceding analysis of bias and anchoring failure in retrieval-augmented generation, this chapter turns to the adversarial threat surface that such judges themselves present. The central tension is that the same evaluative capacity that makes LLM judges useful also renders them exploitable, while every defense that suppresses an attack exacts a measurable cost in benign fidelity, generalization, or deployment overhead. We first characterize the principal attack paradigms and the mechanisms by which they subvert judgment, then examine the defense strategies proposed against them and the trade-offs that constrain each. The chapter closes by tracing how these unresolved tensions motivate the standardization of benchmarking and calibration metrics taken up in the next chapter.

7.1 Adversarial Manipulation of LLM Judges: Attack Surfaces, Bias Amplification, and Failure Modes

What attack vectors compromise the integrity of LLM-as-a-Judge systems, and through what mechanisms do such attacks induce bias and evaluative failure? The evidence reviewed here converges on a sobering picture: the judge is not a passive instrument that merely reports quality but an attack surface whose verdicts can be steered by adversarial perturbations, and whose failure is often systematic rather than random. Three intertwined strands of work—prompt-level injection and jailbreak attacks, adversarial and distribution-shifted inputs, and the endogenous amplification of bias within multi-agent judging architectures—together explain why robustness cannot be assumed from strong baseline safety performance. This subsection synthesizes these strands, contrasts their reported magnitudes, and isolates the boundary conditions under which defenses hold or collapse.

Prompt injection and jailbreak-style manipulation constitute the most direct attack surface, and the evidence indicates that even frontier models are reliably steerable. A comprehensive robustness assessment of LLM-as-a-Judge reports that across seven distinct jailbreak techniques—including refusal suppression, machine translation, prompt injection, reward incentives, and role-playing—most tested judges exhibit pronounced vulnerability, with refusal suppression and machine translation proving the most effective

at degrading safety scores [24]. The magnitude is not marginal: GPT-4o's safety score falls to roughly 0.20 and Claude 3.5 Sonnet to roughly 0.40 after attack, despite both having passed strong baseline safety evaluations [24]. This pattern suggests that safety alignment and judging robustness are partially decoupled capabilities—a model may refuse harmful requests while still being induced to mis-score them. The mechanism is plausibly one of instruction-hierarchy confusion, wherein injected content competes with the judging rubric for control of the output distribution.

A complementary line of evidence reframes the problem statistically rather than mechanistically: judges may fail not because they are successfully attacked but because their verdicts are intrinsically unstable. Work titled around the metaphor of a coin flip finds that LLM judges fail to reliably measure adversarial robustness, with verdicts varying substantially across semantically equivalent presentations of the same input [71]. This is a stronger claim than injection vulnerability, because it implies that reported robustness numbers may be artifacts of judge noise rather than properties of the systems under evaluation. A meta-evaluation of safety judges corroborates this concern, showing that judge robustness itself is rarely meta-evaluated and that ranking stability degrades under perturbation [76]. Read together, these two studies shift the locus of failure from the malicious prompt to the measurement instrument, and they motivate the position that adversarial evaluation of judges must itself be adversarial.

Bias in judging is not only injected from outside but also amplified from within, and multi-agent architectures provide the clearest counterintuitive evidence. A systematic study of multi-agent LLM-as-a-Judge finds that bias consistency across agents increases with debate rounds, indicating that debate functions as a consensus-reinforcing rather than perspective-diversifying mechanism [31]. This directly challenges the intuition that more perspectives yield less prejudice, and it contrasts sharply with single-agent calibration approaches that treat bias as a correctable distributional offset. The same work reports, however, that an LLM-as-meta-judge configuration can outperform debate in stability when the judgment pool is large [31]—an important boundary condition suggesting that aggregation topology, not merely agent count, governs whether bias compounds or cancels.

The mechanistic roots of these failures are beginning to be characterized at the level of internal representations rather than input-output behavior. Interpretability work on judge bias traces unfair verdicts to specific internal features that encode surface attributes of candidates, showing that bias is localized and causally implicated in the scoring decision [7]. This account helps explain why prompt-level defenses may generalize poorly: if bias is mediated by learned representations of spurious attributes, then suppressing a particular injection string leaves the underlying feature pathway intact. It also suggests a division of labor between mechanistic diagnosis and behavioral mitigation, since the former identifies where intervention should occur while the latter demonstrates what can be achieved empirically.

Calibration-based mitigation offers the most quantitatively documented countermeasure, and its reported gains are instructive precisely because they are bounded. CalibraEval reformulates debiasing as an optimization problem and employs a parameter-free order-preserving algorithm to derive a calibration function, reducing selection bias while improving accuracy from 65.27% to 67.13% [29]. The dual improvement—bias reduction without accuracy sacrifice—distinguishes calibration from blunt debiasing heuristics, yet the method's reliance on substantial estimation samples and its limited benefit for already-strong models mark the limits of distributional correction [29]. In contrast, CaLLM pursues bias mitigation through a calibrated judging model rather than post-hoc correction, and a systematic evaluation of bias mitigation strategies across judging pipelines finds that no single strategy dominates across bias types [77]. The comparison indicates that calibration addresses selection bias effectively but does not neutralize injection or adversarial-input attacks, which operate through a different causal pathway.

Defenses aimed specifically at adversarial manipulation adopt a distinct philosophy, treating the judge as a system to be hardened rather than a distribution to be corrected. StableJudge proposes a lightweight adversarial defense framework for LLM-as-a-Judge, targeting the injection channel directly rather than recalibrating outputs. This stands in contrast to calibration and to self-reflection-style refinement, the latter of which improves factual consistency through iterative generate-score-refine loops [42] but was not designed to resist deliberate manipulation. The distinction matters because hallucination mitigation and adversarial robustness are frequently conflated: a judge can become more factually grounded while remaining fully susceptible to a refusal-suppression prefix. Evidence that self-reflection reduces hallucination therefore does not license the inference that it hardens the judge against attack.

The empirical picture is further complicated by conflicting reports about baseline judge competence. Some studies find that LLM judges perform at or near chance on certain adversarial robustness judgments [71], whereas others document meaningful and reproducible discrimination on standard safety benchmarks [76]. These findings are not necessarily contradictory; they are consistent with a task-dependent account in which judges are competent on distributionally familiar judgments and degrade sharply under distribution

shift or adversarial framing. This interpretation is reinforced by the observation that attack effectiveness varies widely across techniques—from near-negligible effects for role-playing and reward incentives to substantial effects for refusal suppression and machine translation [24]—implying that vulnerability is structured by the semantic relationship between the attack and the judging rubric rather than uniformly distributed.

Two gaps in the current evidence deserve emphasis because they constrain how confidently the findings can be generalized. First, the cost and deployability of defenses are rarely compared: calibration requires estimation samples, multi-agent debate incurs substantial inference overhead and carries a risk of problem drift under over-reasoning, and lightweight adversarial defenses have not been benchmarked against these alternatives on common ground [29, 31, 78]. Second, bias metrics remain non-standardized, so reported reductions are difficult to compare across studies [77]. Both gaps are methodological rather than conceptual, and closing them would substantially sharpen the field's ability to adjudicate between competing defenses.

Taken together, the evidence establishes that adversarial failure in LLM judges arises from at least three distinguishable and partially independent causes: instruction-level manipulation of the judging prompt, intrinsic verdict instability under semantically equivalent inputs, and endogenous bias amplification within multi-agent aggregation. These causes are coupled in practice—an injected prompt may exploit the same representational features that produce ordinary bias [7]—which explains why single-mechanism defenses yield incomplete protection. The implication is that robustness claims should be conditioned on the attack family and the aggregation topology, and that evaluation protocols must report both. The following subsection turns from characterizing these vulnerabilities to examining the defense architectures designed to contain them, asking which of the three causal pathways each architecture actually intercepts.

7.2 Intervention Depth and the Pareto Frontier of Judge Hardening

How effective are the mitigation strategies currently proposed for LLM judges—spanning prompt-level defenses, activation steering, calibration, and multi-agent aggregation—at restoring robustness under adversarial pressure? Across the studies reviewed here, the answer is consistent and sobering: no single strategy dominates on robustness, benign fidelity, and deployability simultaneously, and the choice among them is best understood as selecting a point on a Pareto frontier rather than identifying a best method.

The most systematic evidence for this trade-off comes from a unified evaluation of fifteen attacks crossed with eight defenses, which reports that no defense simultaneously achieves the best robustness and benign utility [24]. Re-tokenization attains the lowest residual attack success rate at 16.79%, yet benign utility collapses to 29.67%; conversely, delimiter, sandwich, and instruction-augmentation defenses preserve over 92% benign utility while reducing attack success by only one to four percentage points. A naive LLM detector achieves the lowest composite risk (BDR 30.33%) but incurs an additional judge call per evaluation. The frontier shape is therefore structural rather than incidental: each defense encodes a particular weighting of robustness against utility, and the cost dimension adds a third axis that no single method optimizes jointly.

Steering the judge's internal representations pushes suppression strength furthest, but at a deployment cost that is easy to overlook. Mechanistic analysis locates typed bias directions in the judge's hidden states and applies bidirectional activation steering to Llama-3.1-8B, achieving 38–45% Wasserstein-distance bias reduction under matched rank fidelity (Spearman $\rho \geq 0.85$), compared with only 6–10% for text-rewriting defenses—a four- to six-fold gap that survives five-fold cross-validation with at least 80% of the reduction retained out of sample [7]. The mechanism explains the strength: intervening directly on layer 7–31 activations bypasses the surface form of the prompt entirely. The same mechanism explains the limitation, since optimal intervention layers and strengths vary by an order of magnitude across bias types and the approach presupposes white-box access, leaving closed API judges out of reach.

Calibration-based methods occupy the opposite corner of the frontier, trading coverage for zero fidelity loss. A posterior multi-calibration procedure eliminates detectable miscalibration in 12 of 14 dataset–group pairs and reduces position, verbosity, and family biases by median 64%, 60%, and 46% respectively, against at most 36%, 22%, and 31% for competing baselines, while improving accuracy by 4.6 points and reducing log-loss by 24% [17]. A label-free alternative calibrates the prediction distribution directly, lowering ChatGPT's reward standard deviation from 16.79 to 5.51 and raising accuracy from 65.27% to 67.13% on RewardBench, with a 10% estimation set recovering over 85% of the full-set improvement [29]. Both results indicate that fidelity and debiasing need not conflict—but only for error modes recoverable from output distributions, a scope condition that neither method's reported numbers make salient.

A third design point decouples defense from judge scale altogether. An external 0.13B sentry module

reduces mean absolute score deviation from 1.10 to 0.36 across 298 valid attacks, with 69.5% of defended scores falling within tolerance, while approximately preserving benign performance [78]. This is a materially different bargain from activation steering: the intervention is architectural rather than representational, so it applies to black-box judges, but its validated scope is narrow, centered on score-hijacking injections, and cross-modal coverage remains untested.

Prompt-level debiasing, by contrast, can invert under adversarial conditions, which cautions against treating any single-axis intervention as uniformly beneficial. A systematic comparison finds that position swapping helps on natural evaluation data but degrades Gemini Pro, GPT-4o, and Llama by 4–13 percentage points on the adversarial LLMBar benchmark, while style bias (0.10–0.76) dwarfs position bias (≤ 0.04) in magnitude [77]. The same study reports that the optimal strategy is model-dependent and directional: a combined-budget configuration for Gemini 2.5 Flash reaches 71.0% agreement ($\kappa = 0.549$) at roughly one-fifteenth the cost of the strongest frontier configuration. Debiasing along one axis, in other words, redistributes error to another rather than removing it.

Aggregation-based defenses show similarly conditional returns, and in one important case actively backfire. Multi-agent debate sharpens and amplifies intrinsic bias after the first round and the amplification persists across rounds, whereas an LLM meta-judge resists bias better and sometimes improves with larger judgment pools [31]. The asymmetry is diagnostic: an unbiased proxy agent improves monotonically with debate rounds yet performs below chance when routed through a meta-judge, because the meta-judge cannot recognize unbiased outputs. Adding perspectives therefore does not add independence unless the aggregation topology preserves it—a design constraint rather than a tuning parameter.

Iterative self-correction exhibits the same conditionality in a different modality. A generate–score–refine loop consistently outperforms direct generation across five medical GQA datasets, with up to roughly three-fold gains on MedNLI and reduced human-rated inconsistency, off-topic drift, and factual error [42]. The authors nonetheless concede that the method does not entirely eliminate hallucination and validate only English medical content. The gains are real but bounded by domain and language, and each refinement round adds inference cost—placing this family in the interior of the frontier rather than at its edge.

These per-method comparisons rest on measurements whose reliability is itself contested, which bounds how far any effectiveness claim can be pushed. An audit of 6,442 human-annotated samples finds that state-of-the-art safety judges perform no better than a random coin flip under adversarial distribution shift, that reported attack success rates are systematically inflated, and that best-of-N attacks exploit judge false positives rather than elevating true harmfulness [71]. A complementary meta-evaluation shows that stylistic variation alone can shift the false negative rate by up to 0.24, with some judges misclassifying 100% of harmful generations as safe [76]. If the measurement instrument degrades toward noise under exactly the conditions where defenses are evaluated, then defense efficacy numbers are themselves noisy estimates, and cross-paper comparisons inherit that noise.

Taken together, the evidence supports a three-way constraint rather than a ranking: high bias suppression, high benign fidelity, and high deployability at low cost cannot be jointly attained at current attack strengths. Activation steering leads on suppression but requires white-box access [7]; calibration leads on fidelity but covers only recoverable error modes [17]; detection-based defenses trace the robustness–utility frontier [24]; and aggregation and self-reflection deliver gains only under specific topologies and domains [31, 42]. Practical guidance should therefore be conditioned on deployment context—white-box self-hosted judges favor steering, closed API judges favor calibration, and injection-sensitive scoring favors lightweight external sentries [78]—rather than framed as a single best practice.

Two gaps qualify this conclusion and motivate the analysis that follows. First, almost no study evaluates defenses against an adaptive attacker who re-optimizes against the defense; the one systematic evaluation already shows that optimized inputs with long suffixes can bypass defenses on a production judge [24], suggesting static defense efficacy is likely overstated. Second, validation breadth is thin—most methods are tested on one or two judge families, at most four bias types, and at most three benchmarks—and metric definitions (Cohen's κ, ICC, reward standard deviation) are not comparable across papers. These limitations imply that the frontier reported here is provisional, and they set up the question of what an evaluation protocol capable of certifying defense claims would need to contain.

8 BENCHMARKING AND CALIBRATION STANDARDIZATION OF EVALUATION METRICS

The deficiency that metric definitions are not comparable across the literature advances the question from "how do existing defense lines perform" to "what kind of evaluation protocol do the claims of certification defense lines require," thereby demanding a item-by-item clarification of the constituent elements and

decision conditions that the protocol should possess.

Having established how adversarial pressure exposes vulnerabilities in judging behavior, the review now turns to the measurement apparatus itself. Current practice certifies LLM judges predominantly through accuracy, Pearson correlation, and agreement rate—metrics that capture marginal alignment with a human reference but remain silent on reliability, calibration, robustness to bias, and stability under perturbation, a gap that widens precisely where adversarial conditions make single aggregate scores least trustworthy. Standard dataset validation is positioned as the anchor for credible scoring, while bias mitigation is pursued along two distinct fronts: data-side normalization through lineage transparency, balanced annotation, and gold references that align automatic metrics with human judgment, and judge-side calibration and debiasing that suppress positional, verbosity, stylistic, and self-preference effects. The chapter develops this argument by first interrogating the metric foundations of judge validity, then examining dataset-anchored validation and the separation of data-side from judge-side remedies.

8.1 Consensus Metrics as Necessary-but-Insufficient Evidence of Judge Validity

How far can agreement-based summary statistics—Accuracy, Pearson correlation, and Agreement rate—actually certify that an LLM Judge is effective? The evidence reviewed here converges on a deflationary answer: each of these figures measures only whether a judge's judgments approach some human reference in the marginal distribution, and therefore establishes a necessary but not sufficient condition for validity. This subsection develops that claim along three failure lines—aggregation masking structure, reference dependence, and breakdown under fine-grained or out-of-distribution conditions—and then delimits the scenarios in which each metric remains legitimately usable.

The most direct challenge to correlation-based validation is the decoupling of ranking from scoring. A systematic calibration study of vision-language judges shows that a model can order responses correctly while remaining unable to assign reliable absolute scores, a failure mode that aggregate correlation does not register [9]. On ChartQA the judge attains its highest Pearson correlation among fourteen datasets ($\rho = 0.507$) yet also the widest conformal prediction interval (3.08 on a 1–5 scale); on WIT the correlation falls to $\rho = 0.164$ while the interval narrows to 2.38. A single ρ therefore conflates two qualitatively different judges—one that ranks well but scores poorly, and one that ranks weakly but scores more trustworthily—and the practical implication is that correlation should be read as a statement about ordering, never about calibration.

Aggregate statistics further erase directional structure. The same study reports mean signed errors in which every judge overestimates low-quality responses (ground-truth rating 1: $+1.73$ for one critic model, $+1.98$ for another) and underestimates high-quality ones (rating 5: -0.74 and -1.04 respectively), while the overall bias stays near zero ($+0.38$, $+0.08$) [9]. This compression toward the middle of the scale is a systematic distortion, yet a near-zero bias or a high Pearson coefficient renders it invisible. The mechanism is straightforward: signed errors of opposite direction cancel in the aggregate, so any metric that collapses the error distribution to a single moment is structurally incapable of detecting band-specific miscalibration.

Agreement rate carries an even heavier hidden dependency, namely the quality of the human reference and the judge's own competence. A controlled study varying reference type (none, self, human, wrong, random) establishes that judge–human agreement depends strongly on whether the judge can itself answer the question and whether the supplied reference is correct [36]. Without correct references, agreement drops sharply on questions the judge cannot answer; supplying human-correct references restores most of it, to the point that a 7B model with proper grounding outperforms an ungrounded GPT-4o. Agreement rate thus measures reference quality and task difficulty at least as much as it measures the judge's independent judgment, which is why a reported value is uninterpretable unless paired with a human–human ceiling.

Chance correction does not rescue the metric. A bias-robustness benchmark selects judges using Cohen's $\kappa > 0.8$ as a strong-agreement threshold, yet its confusion analysis reveals that all candidate judges systematically misclassify counter-stereotyped responses as stereotyped [75]. High κ is therefore achievable on the basis of shared blind spots rather than unbiased judgment. Reliability analyses compound the concern: cross-judge κ of 0.51 implies that roughly a quarter of evaluation outcomes depend on which judge is chosen, single evaluations flip 14% of the time, and a single pointwise score carries a 95% error band of ± 1.2 on a ten-point scale [79]. A lone agreement number cannot absorb this noise budget; repeated trials and confidence intervals are required before any such figure can bear evidential weight.

Accuracy exhibits a parallel pathology in fine-grained settings. On the ME benchmark, high question-to-answer accuracy on VCR does not signal genuine understanding: performance on fine-grained sub-questions drops by roughly 20%, and predictions are internally inconsistent, indicating that correct answers may arise from dataset biases and shallow heuristics rather than multimodal comprehension [49]. Accuracy

thus conceals the "right answer for the wrong reason" pattern, since a binary match cannot distinguish grounded inference from shortcut exploitation. The lesson generalizes: whenever the label space is coarse relative to the capability under test, accuracy saturates before understanding does.

Generation-quality metrics show the same fragility under distribution shift. In neural machine translation, BLEU correlates only imperfectly with hallucination: label smoothing raises BLEU by roughly one point while hallucination rates remain comparable, and out-of-distribution test sets exhibit hallucination of 33–35% against 1–2% in-domain, a deterioration BLEU does not reflect [67]. Similarly, CIDEr's correlation with human judgment degrades severely when computed against machine-translated references, and recovers to high Pearson, Spearman, and Kendall values once human gold references are substituted [48]. In both cases the metric itself is unchanged; what changes is the reference distribution, which confirms that reported correlations are partly a function of reference quality rather than of metric capability.

A starker demonstration comes from adversarial comparison tests in text-to-image evaluation. Under the PROTOBIAS protocol, CLIPScore reaches $\kappa \approx 0.001$ and a strong vision-language model judge approaches $\kappa \approx 0$, while human annotators achieve 0.65–0.71 [57]. High pairwise accuracy or high human agreement therefore does not guarantee semantic faithfulness; when the test set is constructed to probe prototypicality bias, automated metrics collapse to chance. This result is best read as a boundary condition: agreement metrics are informative only on distributions that resemble the reference data, and their apparent validity can vanish under targeted perturbation.

These failure lines are not independent but coupled. Reference dependence and aggregation jointly explain why a judge can rank well yet score poorly: the reference supplies the ordering signal, while the absolute scale is never anchored, so signed errors compress toward the middle and cancel in the mean [9, 36]. Shared blind spots, in turn, explain why chance-corrected agreement can be simultaneously high and uninformative [75]. The practical consequence is that no single metric can serve as a diagnostic; calibration intervals, stratified breakdowns, and human baselines must be reported alongside any agreement figure.

The applicable scope of each metric can now be stated precisely. Agreement rate suits low-cost triage of binary or pairwise judgments, provided a human–human baseline is reported as an upper bound [36]. Pearson, Spearman, and Kendall coefficients suit relative comparison of ranking ability and model selection, but not certification of absolute scoring [9]. Accuracy suits classification-style judgments with explicit ground truth, and requires stratified diagnostics in fine-grained, out-of-distribution, or class-imbalanced regimes [49]. All three share a common exclusion zone: absolute scoring, cross-task comparison, and settings with biased or imperfect references.

Two caveats limit the strength of these conclusions. First, the most detailed critiques—ranking–scoring decoupling, the limits of chance correction, and cross-judge noise budgets—derive from a small set of dedicated calibration studies rather than from broad community practice, so the prevalence of these failures remains an open empirical question [79, 57]. Second, metric heterogeneity across subfields—BLEU in translation, CIDEr in captioning, agreement and κ in judge evaluation—means that same-named quantities are often defined differently and must not be compared directly across papers [48, 67]. These caveats suggest that the field's deeper deficit is not a missing metric but an implicit assumption that a single consensus number suffices as evidence of validity.

In sum, Accuracy, Pearson correlation, and Agreement rate each certify only marginal proximity to a reference, and each fails in a characteristic and now well-documented way: aggregation hides directional miscalibration, agreement inherits reference quality and shared bias, and accuracy collapses under fine-grained or shifted distributions. Their legitimate uses are correspondingly narrow, and none extends to absolute scoring or cross-task comparison. This diagnosis motivates the next section, which turns from the limitations of individual metrics to the design of multidimensional reporting standards that combine agreement, calibration intervals, stratified diagnostics, and human baselines into a coherent evaluation protocol.

8.2 Standard-Anchored Validation and Bias Mitigation in Judge Scoring

How can standard datasets be used to validate judge scores and thereby improve their credibility while mitigating human-evaluation bias? The evidence assembled here suggests that no single calibration step suffices: credibility arises from a layered protocol in which human-grounded anchors, distributional calibration, and adversarial stress tests each correct a different failure mode, and the residual gap between machine and human judgment cannot be closed by debiasing alone.

A first and sobering boundary is set by the sample-efficiency ceiling of judge-based evaluation. When the model under evaluation is stronger than the judge, the best achievable bias correction yields an efficiency gain no greater than what one would obtain from roughly twice as much gold-standard data [73]. This

result reframes the role of standard datasets: they are not a legacy artifact to be replaced by scalable judging but the scarce resource that bounds how far judge scores can be trusted. The mechanism is intuitive—a weaker judge cannot resolve distinctions that lie beyond its own competence, so debiasing redistributes error rather than eliminating it. Consequently, any protocol that claims to calibrate judge scores must report its human-anchor budget alongside its agreement statistics, since agreement computed without such anchors may overstate reliability precisely where it matters most.

The dependence of judge–human agreement on the availability of ground truth sharpens this point. Without human reference labels, judge agreement with human judgment drops markedly on questions the judge itself answers incorrectly; supplying human anchors restores most of the lost agreement, and a small model such as Qwen2.5-7B equipped with references can outperform an unreferenced GPT-4o [36]. This finding indicates that the binding constraint is informational rather than parametric: reference labels supply the decision boundary that judges otherwise infer from their own possibly flawed priors. The practical implication is that standard datasets should be curated not merely as test items but as anchor sets whose labels are deliberately distributed across difficulty strata, so that calibration does not collapse exactly on the hard cases where evaluation is most consequential.

Bias profiling of judges reveals which distortions standard validation must target. A systematic evaluation across mitigation strategies finds that style bias dominates, ranging from 0.10 to 0.76, far exceeding position bias at or below 0.04; human raters prefer markdown formatting only 57% of the time whereas four judges do so at rates of 73% to 97%, a systematic misalignment [77]. The asymmetry matters because position bias is largely a within-prompt artifact that swapping or averaging can neutralize, whereas style bias reflects a stable preference learned from pretraining data and therefore survives prompt-level remedies. This suggests that validation sets must be stratified by stylistic perturbation—plain versus formatted, terse versus verbose—if agreement figures are to reflect substance rather than presentation.

Distributional calibration offers a complementary, label-efficient route. CalibraEval calibrates the prediction distribution to counter selection bias without labels, reducing the standard deviation of recall from approximately 11.60, 9.69, and 7.16 on RewardBench, MTBench, and PreferenceBench respectively to an average of 5.04, while raising mean accuracy, for instance from 72.10 to 74.24 for Llama-3-8B [29]. The mechanism is to correct the skew of the judge's output distribution rather than to retrain the judge, which explains why gains appear as variance reduction across benchmarks rather than as uniform accuracy lifts. However, because the method operates without labels, it cannot detect bias that is consistent across the calibration pool; it stabilizes scores without guaranteeing that the stabilized scores are aligned with human standards.

Posterior multi-calibration extends this logic to multiple bias families simultaneously. CaLLM reduces miscalibration significantly in 12 of 14 dataset–bias pairs, with median reductions of roughly 64%, 60%, and 46% for position, verbosity, and family bias respectively, and lifts accuracy on PKU from 0.66 to 0.68 [17]. Compared with single-family calibration, the multi-calibration design acknowledges that biases interact: correcting verbosity alone can leave family bias intact, and the residual two uncorrected pairs indicate that calibration coverage is bounded by the bias taxonomy one is willing to enumerate. The modest accuracy gain alongside large miscalibration reductions suggests that calibration chiefly improves the credibility of score intervals rather than the point estimates themselves.

A distinct threat is that stable scores are not necessarily robust scores. Under post-decision interaction, judge decisions in MT-Bench and AlpacaEval flip 49% of the time under authoritative dialogue challenges, rising to 74% under authoritative prompting, while human agreement falls from 67% to 48% and six of eight ranking entries change position [80]. This dissociation between stability and manipulability is the strongest argument for adversarial validation as a standard component: a judge that reproduces its own scores across repeated runs may still be steered by conversational pressure, and conventional test–retest reliability would not detect it. Validation protocols should therefore include interaction-based perturbations as a separate axis from item-level bias probes.

Architectural remedies for bias have uneven effects, which complicates their adoption as validation standards. Multi-agent debate amplifies intrinsic bias through agent interaction, whereas a meta-judge configuration is more robust; unbiased agents improve steadily across debate rounds yet gain little under the meta-judge [31]. The contrast indicates that adding perspectives is not monotonically beneficial—debate propagates correlated errors among agents sharing pretraining lineage, while the meta-judge aggregates without exposing agents to one another's distortions. For standard-dataset validation, this implies that ensemble-style judges should be validated as distinct systems rather than assumed to inherit the reliability of their components.

Turning from mitigation to verification, dataset-driven judge selection makes credibility auditable. CLEAR-Bias assembles 4,400 prompts plus a human-controlled set and uses Cohen's κ to identify DeepSeek

as the most trustworthy judge, at $\kappa = 0.82$ and Macro-F1 of 0.861, turning judge selection into a checkable dataset-validation procedure [75]. This is a different use of standard data than calibration: rather than adjusting a fixed judge's outputs, it ranks candidate judges against human control labels before deployment. The two approaches are complementary—selection reduces the bias that must later be calibrated—yet selection is only as good as the control set's coverage, and a judge chosen on one prompt distribution may not transfer to another.

Human evaluation itself is not a clean reference, and this qualifies the entire anchoring strategy. Judges exhibit task-dependent uncertainty, ranking reliably while scoring inconsistently in multimodal settings [9], and reliability analyses characterize judge behavior as approaching coin-flip stability on some item types [79]. If the anchor labels carry their own variance, then agreement statistics conflate judge error with reference noise, and the apparent ceiling on judge–human concordance may partly reflect the latter. A credible protocol should therefore report inter-annotator agreement on the anchor set itself, so that the calibration target is a distribution rather than a point.

Taken together, these strands support a layered validation standard rather than a single calibration method. Human anchors bound achievable efficiency [73] and restore agreement where judges are weakest [36]; distributional and posterior calibration reduce variance and miscalibration at low label cost [29, 17]; adversarial and interaction-based probes expose failures invisible to reliability checks [80]; and dataset-driven selection renders judge choice auditable [75]. No layer substitutes for another, and the residual disagreements reported across these studies indicate that bias mitigation shifts error rather than removing it. This composite picture sets up the next section, which asks how such heterogeneous signals—agreement, calibrated intervals, and stratified diagnostics—can be consolidated into reporting standards that remain interpretable to readers who must act on benchmark results.

9 DISCUSSION

Taken together, the four lines of inquiry converge on two cross-cutting threads. The first is an adaptation mismatch: the reliability of an LLM judge is governed less by model scale or benchmark score than by whether the judge's evidence channel, task structure, and evaluation protocol are aligned with the judgment it is asked to make. The second is evaluation fragmentation: the field's dominant metrics, benchmarks, and defense protocols each capture a slice of judge validity while leaving the remainder unmeasured, so that improvements on one axis routinely conceal regressions on another. These two threads are not independent; the adaptation mismatches documented across tasks are precisely what fragmented metrics fail to register, and the fragmentation is what allows mismatched judges to appear adequate.

The need to integrate heterogeneous signals thus translates into the distillation of cross-cutting regularities: when consistency, calibration intervals, and stratified diagnostics are placed within a single analytical framework, two intersecting main threads—adaptive mismatch and evaluation fragmentation—emerge as the common thread that subsumes the various lines of inquiry.

The evidence on modality and evidence provenance sharpens this coupling. Work on multimodal judging shows that visual cues are processed far less reliably than text, that faithfulness and bias can move in opposite directions, and that ostensibly multimodal protocols still rest on text-driven judgment [3, 49, 56]. The retrieval literature supplies the mechanism: judge reliability depends not on the volume of noise but on whether the judge holds an evidence channel independent of the judged output, with plausible-but-wrong substitutes inflating false positives more than outright distractors [72]. Read together, these findings generalize beyond vision: a judge is only as calibrated as the independence of its evidence, and multimodal settings are a special case in which the visual channel is systematically non-independent of the textual narrative. This also qualifies the common assumption that adding modalities adds information; without an independently verifiable channel, added modalities add correlated noise.

The adversarial and calibration results further constrain the metric-level conclusions. Attacks succeed not by breaking judgment wholesale but by exploiting the same shared blind spots that consistency metrics reward, so that high agreement can coexist with systematic misclassification of counter-stereotyped responses [75]. Defenses trade robustness against benign utility and cost, with no single mechanism dominating all three [75], while calibration-oriented approaches can reduce bias substantially without sacrificing fidelity [55, 81]. This directly qualifies the sufficiency claims of agreement- and correlation-based evaluation: a judge can be highly consistent, well calibrated on aggregate, and still fail structurally on the subpopulations that matter [9, 36]. The convergence is that calibration is not a post-hoc correction but a design property that must be evaluated jointly with robustness and utility.

A genuine tension emerges between the retrieval and calibration lines. Retrieval evidence indicates that failures under insufficient evidence can be mitigated but not repaired, with absolute performance

remaining low even after debate-style interventions [72], whereas calibration work reports accuracy gains alongside bias reduction [55, 81]. The reconciliation is that calibration corrects the mapping from evidence to judgment, not the absence of evidence itself; when the gold answer is missing from the retrieved set, no recalibration recovers it. This suggests a division of labor: calibration addresses systematic distortion, retrieval independence addresses evidence sufficiency, and neither substitutes for the other. Similarly, the finding that ranking ability and absolute scoring decouple [9] implies that task structure determines which validity dimension is even measurable, reinforcing the adaptation thread.

The consensus across problems is that single aggregate numbers are necessary but not sufficient evidence of judge validity, and that judge design must be evaluated along at least three separable axes: evidence independence, task-structural fit, and calibration under distributional shift. The unresolved contradiction concerns whether debate and multi-agent protocols mitigate or amplify bias, since the same mechanisms that improve evidence use can entrench shared priors [75, 72]. This remains an open empirical question rather than a settled trade-off.

The reusable contribution of this review is threefold. First, a classification framework that separates judge validity into evidence independence, task-structural fit, and calibration, replacing the single consistency axis that currently organizes the literature. Second, a coordinate system that locates any judge along these dimensions jointly with its robustness-utility-cost profile, making explicit which trade-offs a given design accepts. Third, a design dimension identifying evidence-channel independence as a first-class architectural choice rather than an incidental property of the pipeline. Together these convert the fragmented findings across modalities, retrieval, adversarial settings, and metrics into a common vocabulary for comparing judges and for specifying what a reported number does and does not establish.

10 FUTURE RESEARCH

Future Research

Although the shared vocabulary above enables cross-comparison of evaluation results, it also highlights several unanswered questions; systematically formulating these open questions into testable research directions is precisely the necessary step from comparison toward improvement.

Balanced Perspective: Gaps as Opportunities

The evidence assembled across the four research questions maps a field in which evaluation machinery has outpaced the theory that would justify trusting it. The most conspicuous opening concerns citation accuracy: no included study varies citation correctness as an independent variable and traces the resulting judge performance curve, and even the closest analogue manipulates distractor count rather than citation validity. This is an opportunity rather than a defect, because the same parametric logic that exposes a blind spot can be repurposed to build a graded citation-fidelity benchmark. Complementary work on hallucination causes and lifecycle interventions supplies the conceptual vocabulary for distinguishing fabricated references from misattributed ones, while scholarship-specific analyses show that citation failure is not a generic hallucination but a domain-structured one. A research agenda that treats citation accuracy as a manipulable factor—rather than a post hoc error category—would close a gap that currently limits every downstream claim about scholarly reliability.

Reliability of safety-domain judges constitutes a second, better-evidenced opening. Large-scale human-verified testing indicates that judge accuracy under distribution shift can fall to levels barely distinguishable from chance, and independent bias audits attribute the dominant variance to stylistic rather than positional artefacts. These findings converge on a practical programme: report judge accuracy conditioned on attack type, victim model, and semantic category, and treat style invariance as a first-class metric. The demonstration that lightweight external defenses can reduce mean absolute deviation without touching the judge itself suggests that robustness is partly an architectural question, not only a prompting one. Extending such defenses to the citation dimension is a natural and currently untested next step.

Critical Perspective: Where Evidence Cannot Yet Bear Strong Conclusions

Several directions in the current literature are invoked with more confidence than their evidence supports. Human-gold-referenced studies show that judge–human agreement degrades precisely on items the judge itself answers incorrectly, and that supplying gold references recovers much of the lost agreement. This is a serious caveat for any claim that judges can substitute for annotation at scale, yet it is rarely propagated into benchmark design. Similarly, bias-mitigation evaluations report systematic misalignment between human and judge preferences—human markdown preference near 57% against 73–97% across judges —which undermines the assumption that bias correction restores human-like judgement rather than merely a different bias. Mitigation proposals such as preference-elicitation redesign and adversarial robustness benchmarking for bias elicitation are promising, but neither yet demonstrates transfer beyond

the elicitation format it was designed for.

The citation-accuracy gap must likewise be stated cautiously. Absence of proportional-scan ablations is not proof that citation correctness is unimportant; it is proof that the field has not measured it. The distractor manipulation reported in blind-spot work establishes feasibility of the paradigm but does not license inferences about citation-specific failure modes. Claims that multimodal or concept-level evaluation resolves these issues should be read with equal restraint: LLM-generated concept banks outperform ConceptNet on top-7 concept accuracy and benefit from CLIP-based rather than BERTScore assessment, yet these gains are demonstrated on CIFAR-scale classification and do not automatically transfer to judge reliability in open-ended safety or scholarly settings. Until replication spans domains, strong conclusions about general judge competence remain unsupported.

Synthesised Perspective: Leverage Points Across Problems

Read together, the four questions point to a small number of leverage points that recur across otherwise separate literatures. The first is reference conditioning: providing human gold references restores judge–human agreement even for small models, which can outperform larger unreferenced judges, and the same logic explains why judges fail when key evidence is withheld. Designing evaluation protocols that explicitly control what the judge is shown would simultaneously address reliability, bias, and citation fidelity. The second is external, model-agnostic defense: StableJudge demonstrates that a compact sentry can reduce deviation without modifying the judge, suggesting that robustness investments are better placed in the evaluation harness than in repeated judge fine-tuning.

The third leverage point is measurement validity itself. Hallucination research shows that escape and persistence depend on lifecycle stage rather than on a single model property, and survey work similarly argues for cause-stratified rather than aggregate reporting. Applying the same stratification to judges—by attack, by domain, by evidence availability—would make results comparable across studies that currently report incompatible aggregate numbers. Domain-diverse benchmarks such as comprehensive multimodal medical evaluation illustrate how breadth can be institutionalised, and bias-elicitation benchmarking shows how adversarial stress can be made a standard component rather than an optional extra. A shared protocol combining reference conditioning, external defense, and stratified reporting is the most plausible route from scattered partial findings to cumulative evidence.

11 LIMITATIONS

The evidence base combines an automated retrieval pool with manual supplementation, and this combination bounds what the review can claim to cover. The automated queries favor indexed venues and English-language sources, so work published in other languages, in regional outlets, or outside standard indexes is underrepresented unless it was added by hand. Manual supplementation itself follows the authors' judgment of relevance, which introduces a selection bias that cannot be fully reconstructed after the fact. The literature search closed at a fixed date, and later publications are excluded by construction; where a field moves quickly, the most recent developments may therefore be missing.

Building on the above, incorporating adversarial pressure into standardized protocols offers a workable path to accumulating evidence, but the coverage boundaries and retrieval biases of the underlying evidence base directly limit how far the preceding conclusions can be generalized.

Methodologically, the review uses model-assisted extraction and drafting, and we have made the process as verifiable as possible. Each claim is annotated with the supporting references, and the analytical steps are documented so that readers can trace how conclusions were derived from the retrieved sources. These measures reduce, but do not eliminate, the risk of error: individual extraction mistakes may persist, particularly where source texts are ambiguous, tables are complex, or terminology is inconsistent across studies. Readers should treat the annotations as an audit trail rather than a guarantee of perfect accuracy.

In scope, the review concentrates on the defined topic and admits adjacent areas only where they intersect with it directly. This focus keeps the synthesis coherent but means that neighboring literatures are covered selectively rather than comprehensively. Findings from those adjacent areas are reported only to the extent that they bear on the central question, and the review does not claim to survey them in their own right. Conclusions therefore apply to the topic as framed here, and extending them beyond that frame requires independent justification.

12 CONCLUSION

This review set out to consolidate what is known about LLM-as-a-Judge across four thematic strands, and each strand yields a convergent answer. On evaluation blind spots, the evidence shows that judge performance is markedly weaker on visual than on textual modality, so multimodal judging carries a

systematic bias that hallucination, visual misreading, and cross-modal interaction failure jointly amplify. On bias and anchoring in RAG settings, robustness is governed less by the amount of noise than by whether the judge has a channel through which evidence can be independently verified; context-length limits and retrieval-context dependence further erode consistency. On adversarial vulnerability, attacks expose severe weaknesses, while existing defenses are caught in a three-way constraint among suppression strength, benign fidelity, and deployability, with no single strategy dominating all dimensions. On benchmarking and calibration, accuracy, Pearson correlation, and agreement rate remain the mainstream measures because they map directly onto the question of human alignment, and standardized dataset validation serves as the anchor for credibility while bias mitigation must distinguish data-level from evaluator-level sources.

Having delineated the applicable boundaries of the survey's conclusions, it is natural to ask where these conclusions ultimately converge across the four thematic threads; only by examining, one by one, the direction of the evidence for each thread can we determine whether a consensus sufficient to support judgment has formed within the delineated framework.

Read together, these findings mark a shift in how the judge should be understood: not as a passive scorer of outputs but as an evidence-seeking agent whose reliability is bounded by the verification channels it can access and the modality it must process. Failure, on this view, is not an occasional error to be averaged away but a structured property of the judging setup itself.

For researchers, the practical implication is to treat verification access, modality coverage, and adversarial exposure as first-class design variables in both judge construction and benchmark design, rather than as post hoc robustness checks. For engineers, the implication is to deploy judges only where an independent evidence channel can be instrumented, and to calibrate against standardized datasets before trusting judge scores in multimodal, retrieval-augmented, or adversarial pipelines.

REFERENCES

[1] Fangyi Yu. When ais judge ais: The rise of agent-as-a-judge evaluation for llms. arXiv (search), 2026.

[2] Khaoula Chehbouni, Melina Medjdoub, Florian Carichon, and Golnoosh Farnadi. Llj cards: Best practices for the use of llms as judges. arXiv (Cornell University), 2026.

[3] Aravind Narayanan, Vahid Reza Khazaie, and Shaina Raza. Bias in the picture: Benchmarking vlms with social-cue news images and llm-as-judge assessment. arXiv (search), 2026.

[4] Zewen Liu. Multimodal evaluator preference collapse: Cross-modal coupling in self-evolving agents. arXiv (Cornell University), 2026.

[5] Yiyang Huang, Yitian Zhang, Yizhou Wang, and Mingyuan Zhang. Distorted or fabricated? a survey on hallucination in video llms. arXiv (search), 2026.

[6] Nitish Kumar, Prateik Pothuneedi, and Greg Bandak. The anatomy of a blind spot: What an llm judge must be shown, and how much. arXiv (search), 2026.

[7] Zixiang Xu, Sixian Li, Huaxing Liu, and Xiang Wang. Inside the unfair judge: A mechanistic interpretability account of llm-as-judge bias. arXiv (Cornell University), 2026.

[8] Riya Tapwal, Abhishek Kumar, and Carsten Maple. Faithful or fabricated? a causal framework for rationalization bias in llm judges. arXiv (search), 2026.

[9] Divake Kumar, Sina Tayebati, Devashri Naik, and Ranganath Krishnan. Vlm judges can rank but cannot score: Task-dependent uncertainty in multimodal evaluation. arXiv (Cornell University), 2026.

[10] Bo Pan, Xuan Kan, Kaitai Zhang, and Yan Yan. Bi-level prompt optimization for multimodal llm-as-a-judge. arXiv (search), 2026.

[11] Xiangru Jian, Hao Xu, Wei Pang, and Xinjian Zhao. Forge: Fine-grained multimodal evaluation for manufacturing scenarios. arXiv (Cornell University), 2026.

[12] Yusheng He, Jizhe Zhou, Xia Du, and Zheng Lin. What makes lvlms hallucinate less? unveiling the architectural factors behind hallucination robustness. arXiv (search), 2026.

[13] Xianya Fang, Feiyang Ren, Xiang Chen, and Yu Tian. Beyond superficial unlearning: Sharpness-aware robust erasure of hallucinations in multimodal llms. arXiv (search), 2026.

[14] Alex Robertson, Huizhi Liang, Mahbub Gani, and Rohit Kumar. Kghalubench: A knowledge graph-based hallucination benchmark for evaluating the breadth and depth of llm knowledge. arXiv (search), 2026.

[15] Songeun Chae, Min Kim, Donghoon Jung, and Seojin Choi. Self- and other-labels induce bidirectional bias in llm judges. arXiv (search), 2026.

[16] Benjamin Feuer, Lucas Rosenblatt, and Oussama Elachqar. Towards provably unbiased llm judges via bias-bounded evaluation. arXiv (search), 2026.

[17] Cesare Barbera, Luca Stradiotti, Lorenzo Cascioli, and Lorenzo Perini. Callm: A callm judge is a better judge: Mitigating bias in llm-as-a-judge. arXiv (search), 2026.

[18] Abhigya Verma, Amit Kumar Saha, Seganrasan Subramanian, and Sai Harshitha Aluru. Agent-judgebench: A multi-difficulty benchmark for evaluating llm judges on agentic tool-calling. arXiv (search), 2026.

[19] Shreyas KC. Babeljudge: Measuring llm-as-a-judge reliability across languages and agent trajectories. arXiv (search), 2026.

[20] Pengcheng Chen, Jin Ye, Guoan Wang, and Yanjun Li. Gmai-mmbench: A comprehensive multimodal evaluation benchmark towards general medical ai. neural information processing systems, 2024.

[21] Jeffrey Flynt. Groundeval: A deterministic replacement for llm-as-judge in stateful agent evaluation. arXiv (Cornell University), 2026.

[22] Hippolyte Gisserot-Boukhlef, Nicolas Boizard, Emmanuel Malherbe, and Céline Hudelot. Bert-as-a-judge: A robust alternative to lexical methods for efficient reference-based llm evaluation. arXiv (search), 2026.

[23] Hui Wei, Shenghua He, Tian Xia, and Fei Liu. Systematic evaluation of llm-as-a-judge in llm alignment tasks. arXiv (search), 2026.

[24] Songze Li, Chuokun Xu, Jiaying Wang, and Xueluan Gong. Llms cannot reliably judge (yet?): A comprehensive assessment on the robustness of llm-as-a-judge. arXiv (search), 2026.

[25] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, and Siyuan Zhuang. Judging llm-as-a-judge with mt-bench and chatbot arena. arXiv (search), 2026.

[26] Association for Computational Linguistics 2025, Badshah, Sher, Sajjad, and Hassan. Reference-guided verdict: Llms-as-judges in automatic evaluation of free-form qa, 2026. Preprint.

[27] Xinran Zhang. Atomic and holistic llm judges for reference-grounded support labels: A prompt-controlled comparison. arXiv (search), 2026.

[28] Yuqi Tang, Kehua Feng, Yunfeng Wang, and Zhiwen Chen. Learning an efficient multi-turn dialogue evaluator from multiple llm judges. arXiv (search), 2026.

[29] Haitao Li, Junjie Chen, Qingyao Ai, Zhumin Chu, Yujia Zhou, Qian Dong, and Yiqun Liu. Calibraeval: Calibrating prediction distribution to mitigate selection bias in llms-as-judges, 2026. Preprint.

[30] Yoshinari Fujinuma. Contrastive decoding mitigates score range bias in llm-as-a-judge. arXiv (search), 2026.

[31] Chiyu Ma, Enpei Zhang, Yilun Zhao, Wenjun Liu, Yaning Jia, Peijun Qing, Lin Shi, Arman Cohan, Yujun Yan, and Soroush Vosoughi. Judging with many minds: Do more perspectives mean less prejudice? on bias amplification and resistance in multi-agent based llm-as-judge, 2026. Preprint.

[32] Arash Marioriyad, Mohammad Hossein Rohban, and Mahdieh Soleymani Baghshah. The silent judge: Unacknowledged shortcut bias in llm-as-a-judge. arXiv (search), 2026.

[33] Dongryeol Lee, Yerin Hwang, Yongil Kim, and Joonsuk Park. Are llm-judges robust to expressions of uncertainty? investigating the effect of epistemic markers on llm-based evaluation. arXiv (search), 2026.

[34] Yerin Hwang, Dongryeol Lee, Taegwan Kang, and Yongil Kim. Can you trick the grader? adversarial persuasion of llm judges. arXiv (search), 2026.

[35] Sunishchal Dev, Andrew Sloan, Joshua Kavner, and Nicholas Kong. Judge reliability harness: Stress testing the reliability of llm judges. arXiv (search), 2026.

[36] Michael Krumdick, Charles Lovering, Varshini Reddy, and Seth Ebner. No free labels: Limitations of llm-as-a-judge without human grounding. arXiv (search), 2026.

[37] Yuanning Feng, Sinan Wang, Zhengxiang Cheng, and Yao Wan. Are we on the right way to assessing llm-as-a-judge? arXiv (search), 2026.

[38] Dongryeol Lee, Yerin Hwang, Taegwan Kang, and Minwoo Lee. Judging against the reference: Uncovering knowledge-driven failures in llm-judges on qa evaluation. arXiv (search), 2026.

[39] Naveen Lamba, Sanju Tiwari, and Manas Gaur. Hallucinations in llms: A lifecycle-based survey of causes, detection, mitigation, and prevention, 2026. Preprint.

[40] Sourav Banerjee, Ayushi Agarwal, and Saloni Singla. Llms will always hallucinate, and we need to live with this. arXiv (search), 2026.

[41] Miriam Rateike, Celia Cintas, John Wamburu, and Tanya Akumu. Weakly supervised detection of hallucinations in llm activations. arXiv (search), 2026.

[42] Towards mitigating hallucination in large language models via self-reflection, 2026. Preprint.

[43] Hieu Nguyen, Zihao He, Shoumik Atul Gandre, and Ujjwal Pasupulety. Smoothing out hallucinations: Mitigating llm hallucination with smoothed knowledge distillation. arXiv (search), 2026.

[44] Hanxing Ding, Liang Pang, Zihao Wei, and Huawei Shen. Rowen: Adaptive retrieval-augmented generation for hallucination mitigation in llms. arXiv (search), 2026.

[45] Abd Elrahman Amer and Magdi Amer. Using multi-agent architecture to mitigate the risk of llm hallucinations. arXiv (search), 2025.

[46] Demian Till, John Smeaton, Peter Haubrick, and Gouse Saheb. Teaming llms to detect and mitigate hallucinations. arXiv (search), 2026.

[47] Weichen Zhang, Yiyou Sun, Pohao Huang, and Jiayue Pu. Mirage-bench: Llm agent is hallucinating and where to find them. arXiv (search), 2026.

[48] Ashish V. Thapliyal, Jordi Pont-Tuset, Xi Chen, and Radu Soricut. Crossmodal-3600: A massively multilingual multimodal evaluation dataset, 2026. Preprint.

[49] Zhecan Wang, Haoxuan You, Yicheng He, Wenhao Li, Kai-Wei Chang, and Shih-Fu Chang. Understanding me? multimodal evaluation for fine-grained visual commonsense, 2026. Preprint.

[50] Jirayu Burapacheep, Ishan Gaur, Agam Bhatia, and Tristan Thrush. Colorswap: A color and word order dataset for multimodal evaluation, 2026. Preprint.

[51] Jiacheng Chen, Tianhao Liang, Sherman Siu, and Zhengqing Wang. Mega-bench: Scaling multimodal evaluation to over 500 real-world tasks. arXiv (Cornell University), 2024.

[52] Botian Jiang, Lei Li, Xiaonan Li, and Zhaowei Li. Understanding the role of llms in multimodal evaluation benchmarks. arXiv (Cornell University), 2026.

[53] Yujia Xiao, Liumeng Xue, Lei He, and Xinyi Chen. Podeval: A multimodal evaluation framework for podcast audio generation. arXiv (Cornell University), 2026.

[54] Yuan Si, Simeng Han, Daming Li, and Hanyuan Shi. Scratcheval: A multimodal evaluation framework for llms in block-based programming. arXiv (Cornell University), 2026.

[55] Jian Weng, Jia-Wei Liao, Cheng-Fu Chou, and Jun-Cheng Chen. M-erasurebench: A comprehensive multimodal evaluation benchmark for concept erasure in diffusion models, 2026. Preprint.

[56] Jongwoo Ko, Sungnyun Kim, Sungwoo Cho, and Se-Young Yun. Flex-judge: Text-only reasoning unleashes zero-shot multimodal evaluators. arXiv (search), 2026.

[57] Subhadeep Roy, Gagan Bhatia, and Steffen Eger. Prototypicality bias reveals blindspots in multimodal evaluation metrics. arXiv (Cornell University), 2026.

[58] Yunna Cai, Fan Wang, H. Wang, and Kun Wang. Exploring safety alignment evaluation of llms in chinese mental health dialogues via llm-as-judge. arXiv (Cornell University), 2026.

[59] Jiaxin Gao, Chen Chen, Yanwen Jia, and Xueluan Gong. Evaluating and mitigating llm-as-a-judge bias in communication systems. arXiv (search), 2026.

[60] Jennifer D'Souza, Hamed Babaei Giglou, and Quentin Münch. Yescieval: Robust llm-as-a-judge for scientific question answering. arXiv (search), 2026.

[61] Kyungho Bae, Jinhyung Kim, Sihaeng Lee, and Soonyoung Lee. Mash-vlm: Mitigating action-scene hallucination in video-llms through disentangled spatial-temporal representations. arXiv (search), 2026.

[62] Chung-En, Yu, Hsuan-Chih, and Chen. Hydra: An agentic reasoning approach for enhancing adversarial robustness and mitigating hallucinations in vision-language models. arXiv (search), 2026.

[63] Yasser Hamidullah, Koel Dutta Chowdhury, Yusser Al Ghussin, and Shakib Yazdani. Grounding or guessing? visual signals for detecting hallucination in sign language translation. arXiv (search), 2026.

[64] Dong-Xiao Zhang, Hu Lou, Jun-Jie Zhang, and Jun Zhu. Neural uncertainty principle: A unified view of adversarial fragility and llm hallucination. arXiv (search), 2026.

[65] Chengxu Yang, Jingling Yuan, Siqi Cai, and Jiawei Jiang. Heaven-sent or hell-bent? benchmarking the intelligence and defectiveness of llm hallucinations. arXiv (search), 2026.

[66] Yejin Bang, Ziwei Ji, Alan Schelten, and Anthony Hartshorn. Hallulens: Llm hallucination benchmark. arXiv (search), 2026.

[67] Chaojun Wang and Rico Sennrich. On exposure bias, hallucination and domain shift in neural machine translation, 2026. Preprint.

[68] Harsh Mehta, Yoav Artzi, Jason Baldridge, and Eugene Ie. Retouchdown: Adding touchdown to streetlearn as a shareable resource for language grounding tasks in street view. arXiv (OAI-PMH), 2026.

[69] Unveiling ethnic tourism imaginaries: A multimodal evaluation, 2026. Preprint.

[70] Jiawei Gu, Jiang, Xuhui, Zhichao Shi, Hexiang Tan, Zhai, Xuehao, Xu, Chengjin, Wei Li, Yinghan Shen, and Shengjie Ma. A survey on llm-as-a-judge, 2026. Preprint.

[71] Leo Schwinn, Moritz Ladenburger, Tim Beyer, and Mehrnaz Mofakhami. A coin flip for safety: Llm judges fail to reliably measure adversarial robustness. arXiv (search), 2026.

[72] Wentao Hu, Wengyu Zhang, Yiyang Jiang, Chen Zhang, Xiaoyong Wei, and Qing Li. Removal of hallucination on hallucination: Debate-augmented rag, 2026. Preprint.

[73] Florian E. Dorner, Vivian Y. Nastl, and Moritz Hardt. Limits to scalable evaluation at the frontier: Llm as judge won't beat twice the data. arXiv (Cornell University), 2026.

[74] Chalamalasetti Kranti and Sowmya Vajjala. Llm judges can be too generous when there is no reference answer. arXiv (search), 2026.

[75] Riccardo Cantini, Alessio Orsino, Massimo Ruggiero, and Domenico Talia. Benchmarking adversarial robustness to bias elicitation in large language models: Scalable automated assessment with llm-as-a-judge, 2026. Preprint.

[76] Francisco Eiras, Eliott Zemour, Eric Lin, and Vaikkunth Mugunthan. Know thy judge: On the robustness meta-evaluation of llm safety judges. arXiv (Cornell University), 2026.

[77] Sadman Kabir Soumik. Judging the judges: A systematic evaluation of bias mitigation strategies in llm-as-a-judge pipelines. arXiv (search), 2026.

[78] Anning Hu, Yangyu Li, Yuqi Huang, and Shengchao Hu. Stablejudge: A lightweight adversarial defense framework for llm-as-a-judge. arXiv (search), 2026.

[79] Abel Yagubyan. The coin flip judge? reliability and bias in llm-as-a-judge evaluation. arXiv (search), 2026.

[80] Srimonti Dutta and Akshata Kishore Moharir. Stability vs. manipulability: Evaluating robustness under post-decision interaction in llm judges. arXiv (search), 2026.

[81] Usman Ahmad and Vladimir Vladimirovich Ivanov. Automating high-quality concept banks: leveraging llms and multimodal evaluation metrics, 2026. Preprint.